



Project Report

**Title: Development of a Multi-Campus Subnet-Based University
Network Architecture**

Course Details

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Submitted To

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Table of Contents

1. Problem Statement.....	3
2. Network Design	3
2.1 Overall Architecture.....	3
2.2 IP Addressing Scheme	4
2.4 Static Routing Configuration.....	15
2.5 DHCP Server Configuration	19
2.6 DNS Server Configuration.....	30
2.7 Web Server Configuration.....	32
3. Experimental Results	33
3.1 Ping Test Results	33
3.2 Web Access Verification	36
4. Conclusion	40

1. Problem Statement

In the modern digital era, universities depend heavily on reliable computer networks to manage academic, administrative and online services efficiently. A multi campus university such as Stamford University operates across multiple locations and relies on a stable and secure network infrastructure to support essential services including admissions, advising, result processing, eTender systems, library management, accounts and continuous access to the university website.

The primary challenge addressed in this project is the design and implementation of a scalable, fully connected multi-campus network consisting of five campuses: AB1, AB2, AB3, AB4 and AB5. Each campus must support wired hosts, automatic IP address assignment, reliable inter campus data communication and access to centralized services such as a DNS server and a university web server.

The network is designed using Class C IP addressing with proper subnetting, static routing and distributed DHCP servers, ensuring efficient and reliable communication among all campuses. Additionally, users from any campus should be able to access the university website using a domain name rather than an IP address, improving usability and overall network efficiency.

2. Network Design

2.1 Overall Architecture

The complete network is divided into five separate campus LANs (AB1 to AB5). Each campus includes one router, one switch, four wired PCs, and one dedicated DHCP server to ensure automatic IP address allocation within that campus.

All campus routers are interconnected through serial WAN point-to-point links, forming a unified university-wide backbone network. Static routing is implemented to establish reliable connectivity between all routers. A central DNS Server and Web Server are placed in Campus AB1, enabling domain-based access to the university website from every campus. This architecture ensures efficient traffic flow, easier troubleshooting, and supports future scalability.

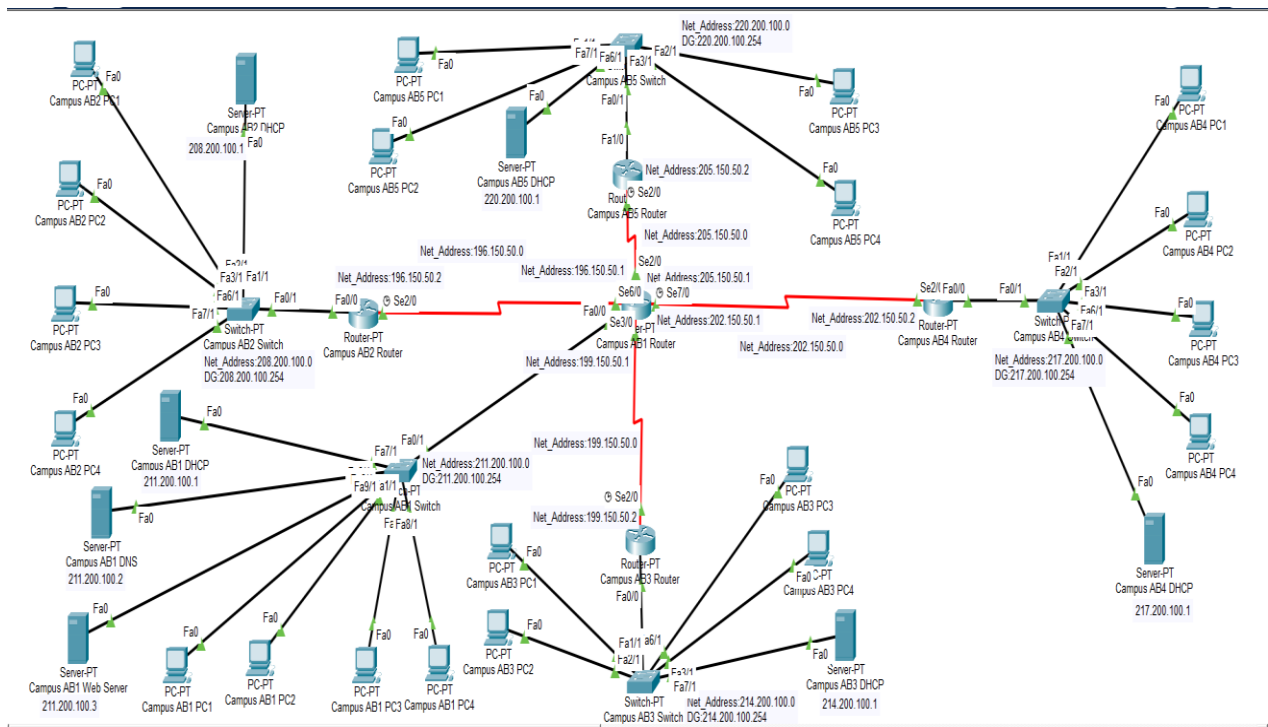


Figure 1: A Complete Multi Campus Network Topology

2.2 IP Addressing Scheme

All IP addresses used in this network follow a Class C addressing scheme. To ensure proper network organization and efficient traffic management, separate subnets are allocated for each campus LAN and for each WAN serial DCE link connecting the routers.

Link	Network Address	Router Interface IPs
AB1 – AB2	196.150.50.0	196.150.50.1 and 196.150.50.2
AB1 – AB3	199.150.50.0	199.150.50.1 and 199.150.50.2
AB1 – AB4	202.150.50.0	202.150.50.1 and 202.150.50.2
AB1 – AB5	205.150.50.0	205.150.50.1 and 205.150.50.2

Table 1: WAN Serial Link Networks

Campus	Network Address	Subnet Mask	Default Gateway	DHCP Server IP
AB1	211.200.100.0	255.255.255.0	211.200.100.254	211.200.100.1
AB2	208.200.100.0	255.255.255.0	208.200.100.254	208.200.100.1
AB3	214.200.100.0	255.255.255.0	214.200.100.254	214.200.100.1
AB4	217.200.100.0	255.255.255.0	217.200.100.254	217.200.100.1
AB5	220.200.100.0	255.255.255.0	220.200.100.254	220.200.100.1

Table: IP Address Allocation for Campus LANs (Switch Based Networks)

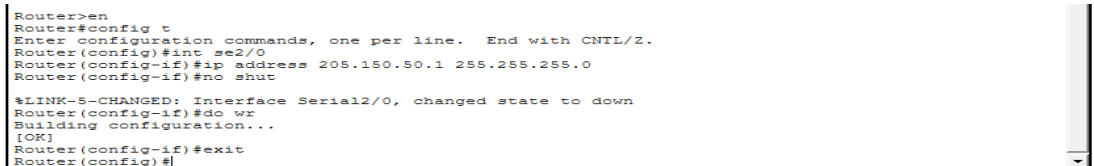
2.3 Router Interface Configuration

The first step of the network implementation involved configuring all LAN and WAN interfaces using Class C IP addresses. FastEthernet interfaces were assigned Class C IP addresses to function as default gateways for the LANs, while Serial interfaces were also configured with Class C addressing to enable WAN communication between routers. Serial DCE interfaces were set with appropriate clock rates to ensure proper synchronization. This configuration established the basic connectivity of the entire network.

➤ Router Configurations for Campus AB1

Se2/0

```
Router>en
Router#config t.
Router(config)#int se2/0
Router(config-if)#ip address 205.150.50.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```



```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 205.150.50.1 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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Se 3/0

```
Router(config-if)#exit
Router(config)#int se3/0
Router(config-if)#ip address 199.150.50.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int se3/0
Router(config-if)#ip address 199.150.50.1 255.255.255.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
```

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Se6/0

```
Router(config)#int se6/0
Router(config-if)#ip address 196.150.50.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int se6/0
Router(config-if)#ip address 196.150.50.1 255.255.255.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial6/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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Se7/0

```
Router(config)#int se7/0
Router(config-if)#ip address 202.150.50.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int se7/0
Router(config-if)#ip address 202.150.50.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial7/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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Fa0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 211.200.100.254 255.255.255.0
Router(config-if)#no shut
```

```
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int fa0/0
Router(config-if)#ip address 211.200.100.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
Router(config)#
```

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```
Campus AB1 Router
Physical Config CLI Attributes
IOS Command Line Interface

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se6/0
Router(config-if)#ip address 196.150.50.1 255.255.255.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial6/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se7/0
Router(config-if)#ip address 202.150.50.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial7/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 211.200.100.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial6/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial6/0, changed state to up
```

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Figure 2: Router Interface Configuration (CLI) for AB1

➤ Router Configurations for Campus AB2

Se2/0

```
Router>en
Router#config t
```

```
Router(config)#int se2/0
Router(config-if)#ip address 196.150.50.2 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 196.150.50.2 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
```

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Fa0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 208.200.100.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int fa0/0
Router(config-if)#ip address 208.200.100.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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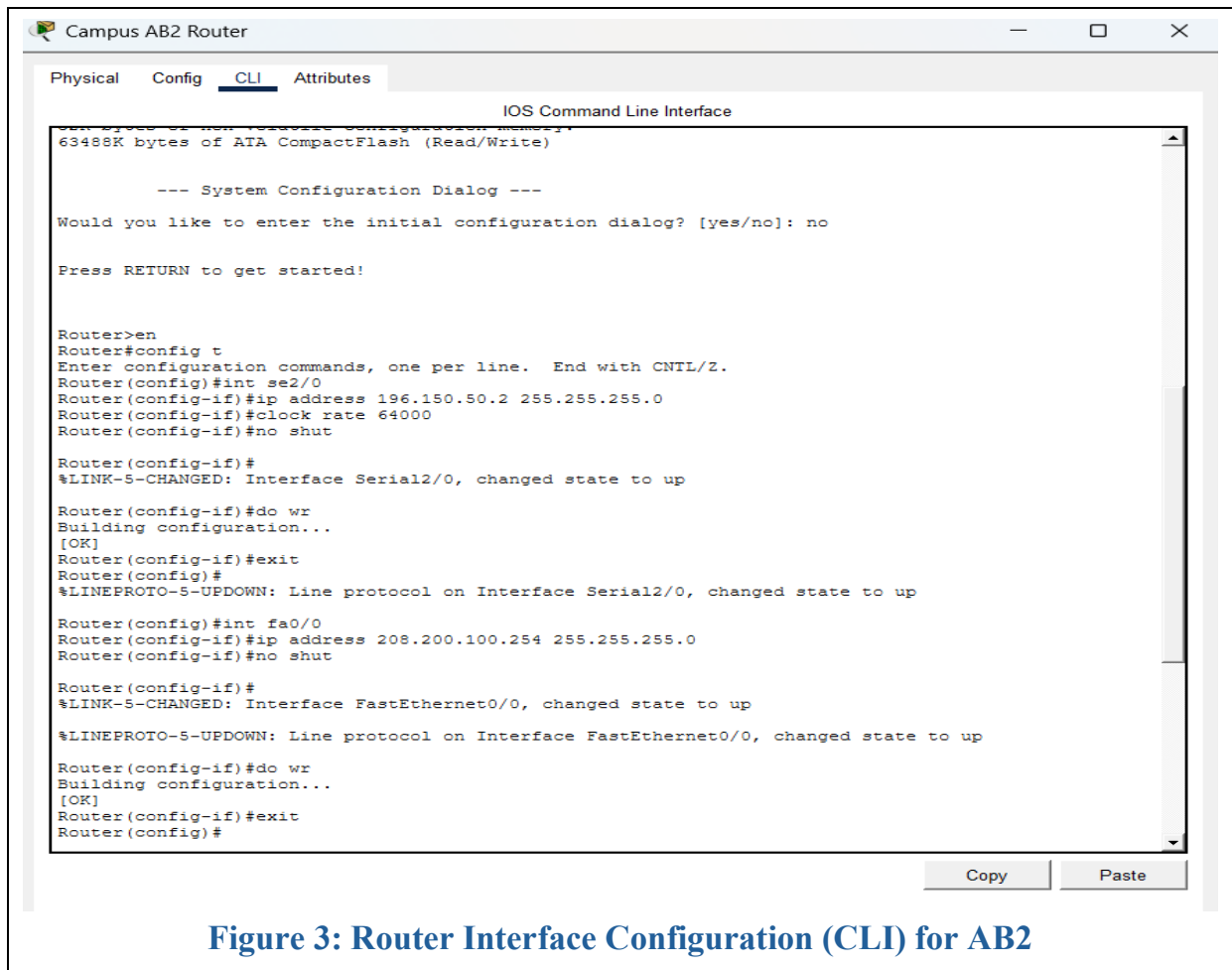


Figure 3: Router Interface Configuration (CLI) for AB2

➤ Router Configurations for Campus AB3

Se2/0

```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 199.150.50.2 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 199.150.50.2 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config)#
```

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Fa0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 214.200.100.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int fa0/0
Router(config-if)#ip address 214.200.100.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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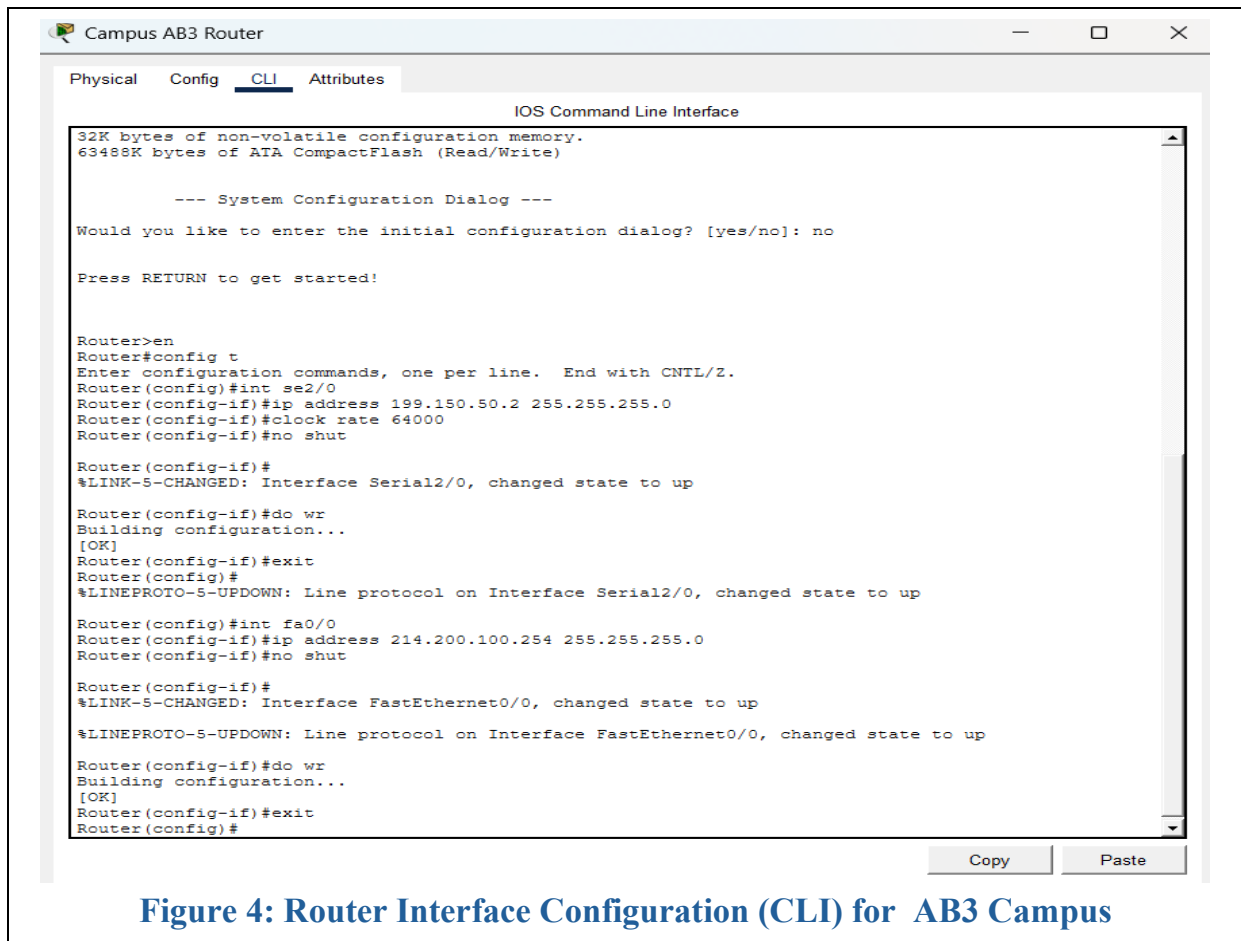


Figure 4: Router Interface Configuration (CLI) for AB3 Campus

➤ Router Configurations for Campus AB4

Se2/0

```
Router>en
Router#config t
Router(config)#int se 2/0
Router(config-if)#ip address 202.150.50.2 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 202.150.50.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config)#
```

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Fa0/0

Router(config)#int fa0/0

Router(config-if)#ip address 217.200.100.254 255.255.255.0

Router(config-if)#no shut

Router(config-if)#do wr

Router(config-if)#exit

```
Router(config)#int fa0/0
Router(config-if)#ip address 217.200.100.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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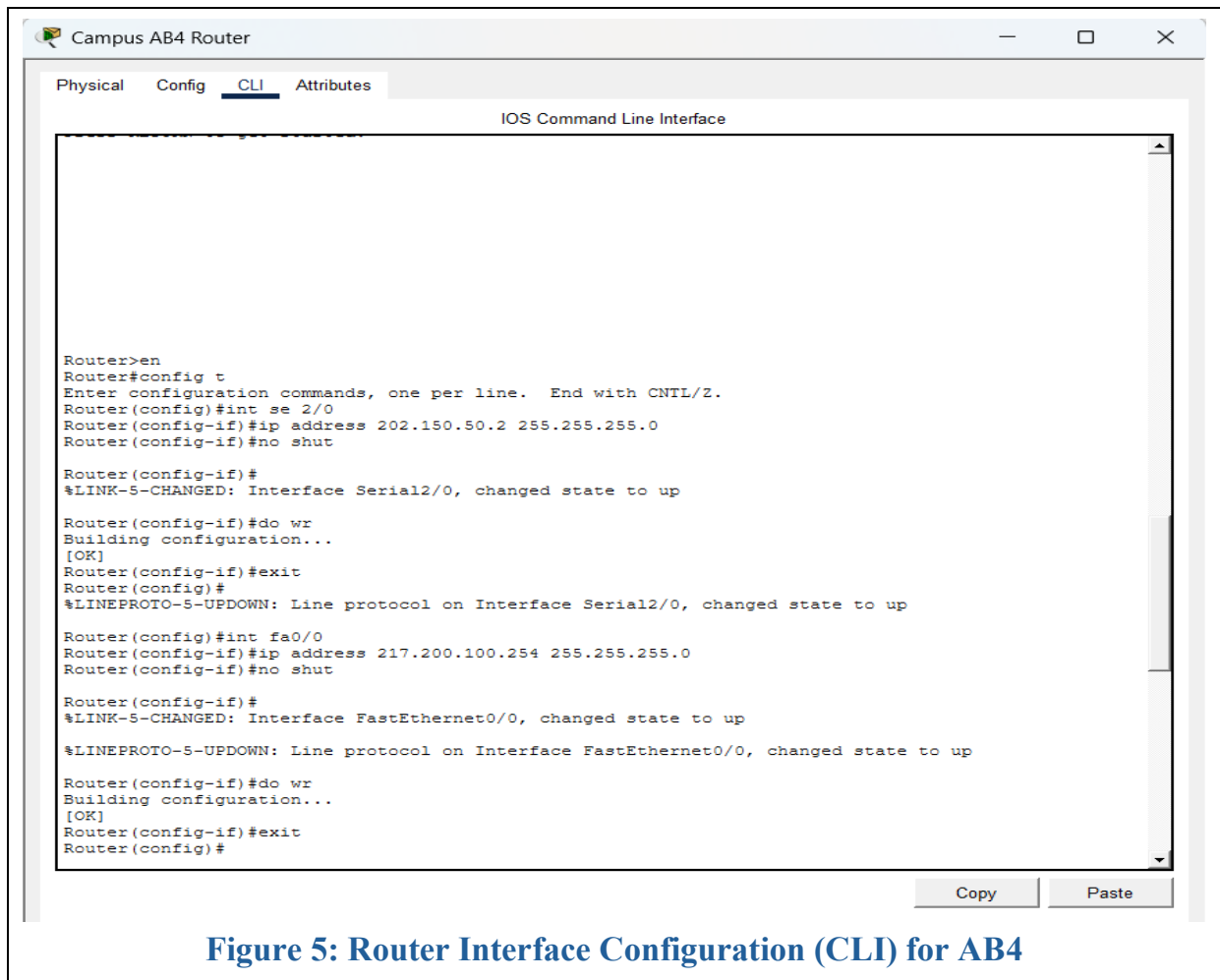


Figure 5: Router Interface Configuration (CLI) for AB4

➤ Router Configurations for Campus AB5

Se2/0

```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 205.150.50.2 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 205.150.50.2 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config)#
```

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Fa1/0

```
Router(config)#int fa1/0
Router(config-if)#ip address 220.200.100.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int fa1/0
Router(config-if)#ip address 220.200.100.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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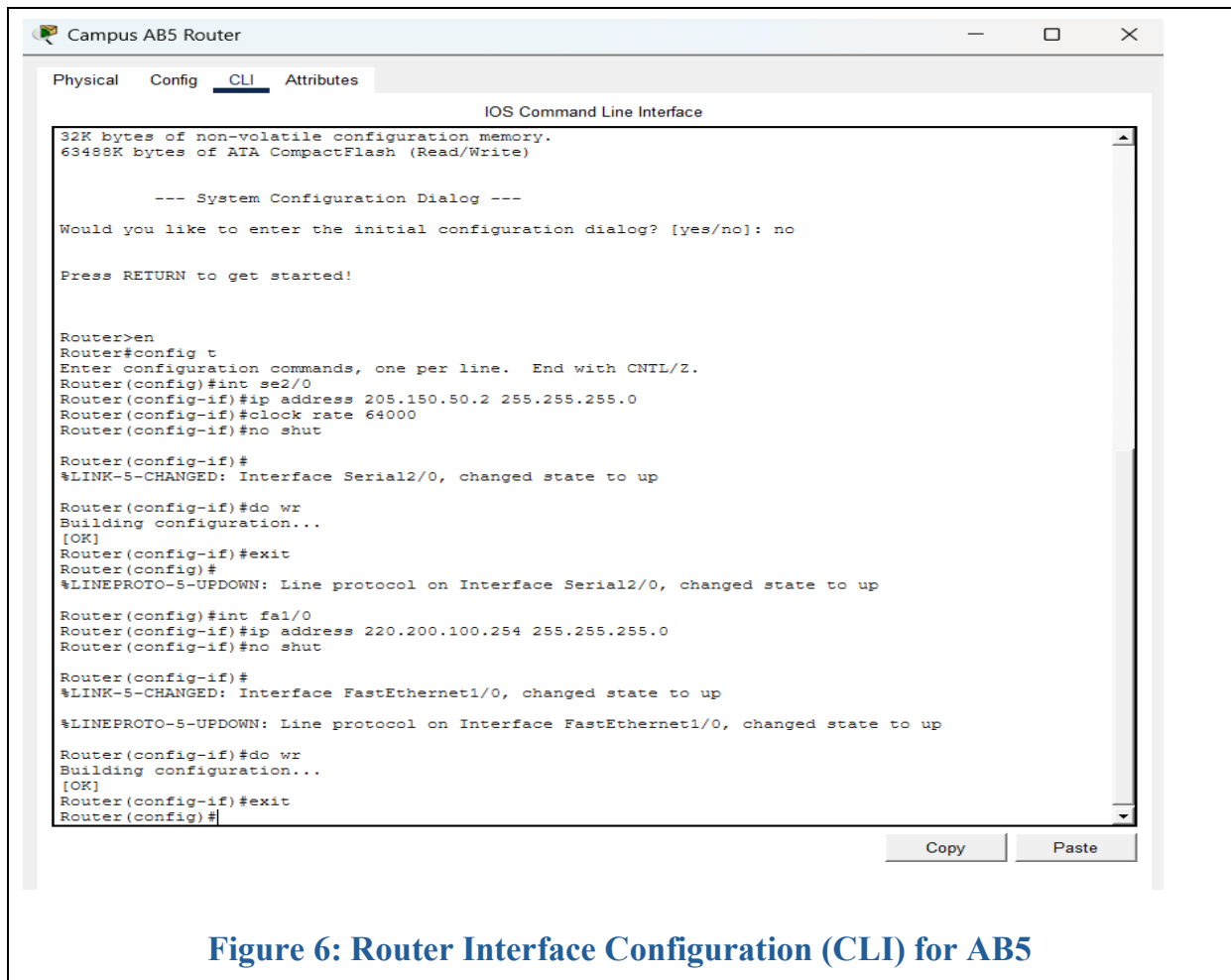


Figure 6: Router Interface Configuration (CLI) for AB5

2.4 Static Routing Configuration

After configuring all router interfaces, static routing was implemented to establish communication between the five campus networks. Static routes were manually added to each router so that it can correctly forward packets to all remote campus LANs. Each routing entry specifies the destination network, subnet mask and next hop IP address.

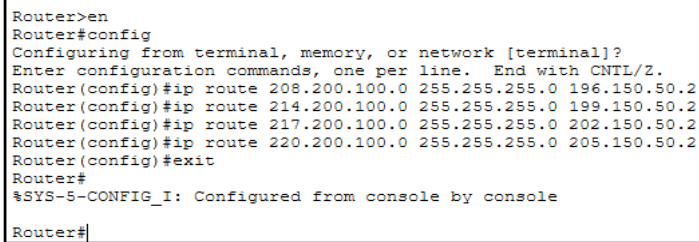
Static routing was chosen for this network design because it offers:

- Complete control over routing decisions
- Predictable and consistent packet forwarding
- Simplicity and transparency, making it suitable for academic analysis and demonstration

This routing approach ensures reliable inter campus connectivity while keeping the network configuration easy to understand, manage and troubleshoot.

➤ Static Routing Protocol for AB1 Campus Router

```
Router>en
Router#config
Router(config)#ip route 208.200.100.0 255.255.255.0 196.150.50.2
Router(config)#ip route 214.200.100.0 255.255.255.0 199.150.50.2
Router(config)#ip route 217.200.100.0 255.255.255.0 202.150.50.2
Router(config)#ip route 220.200.100.0 255.255.255.0 205.150.50.2
Router(config)#exit
```



```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 208.200.100.0 255.255.255.0 196.150.50.2
Router(config)#ip route 214.200.100.0 255.255.255.0 199.150.50.2
Router(config)#ip route 217.200.100.0 255.255.255.0 202.150.50.2
Router(config)#ip route 220.200.100.0 255.255.255.0 205.150.50.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
```

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Figure 7: Static Router Configuration for AB1 Campus

➤ Static Routing Protocol for AB2 Campus Router

```
Router>en
Router#config
Router(config)#ip route 199.150.50.0 255.255.255.0 196.150.50.1
Router(config)#ip route 202.150.50.0 255.255.255.0 196.150.50.1
Router(config)#ip route 205.150.50.0 255.255.255.0 196.150.50.1
Router(config)#ip route 211.200.100.0 255.255.255.0 196.150.50.1
Router(config)#ip route 214.200.100.0 255.255.255.0 196.150.50.1
Router(config)#ip route 217.200.100.0 255.255.255.0 196.150.50.1
Router(config)#ip route 220.200.100.0 255.255.255.0 196.150.50.1
Router(config)#exit
```

```

Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 199.150.50.0 255.255.255.0 196.150.50.1
Router(config)#ip route 202.150.50.0 255.255.255.0 196.150.50.1
Router(config)#ip route 205.150.50.0 255.255.255.0 196.150.50.1
Router(config)#ip route 211.200.100.0 255.255.255.0 196.150.50.1
Router(config)#ip route 214.200.100.0 255.255.255.0 196.150.50.1
Router(config)#ip route 217.200.100.0 255.255.255.0 196.150.50.1
Router(config)#ip route 220.200.100.0 255.255.255.0 196.150.50.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#

```

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Figure 8: Static Router Configuration for AB2 Campus

➤ Static Routing Protocol for AB3 Campus Router

```

Router>en
Router#config
Router(config)#ip route 196.150.50.0 255.255.255.0 199.150.50.1
Router(config)#ip route 202.150.50.0 255.255.255.0 199.150.50.1
Router(config)#ip route 205.150.50.0 255.255.255.0 199.150.50.1
Router(config)#ip route 208.200.100.0 255.255.255.0 199.150.50.1
Router(config)#ip route 211.200.100.0 255.255.255.0 199.150.50.1
Router(config)#ip route 217.200.100.0 255.255.255.0 199.150.50.1
Router(config)#ip route 220.200.100.0 255.255.255.0 199.150.50.1
Router(config)#exit

```

```

Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 196.150.50.0 255.255.255.0 199.150.50.1
Router(config)#ip route 202.150.50.0 255.255.255.0 199.150.50.1
Router(config)#ip route 205.150.50.0 255.255.255.0 199.150.50.1
Router(config)#ip route 208.200.100.0 255.255.255.0 199.150.50.1
Router(config)#ip route 211.200.100.0 255.255.255.0 199.150.50.1
Router(config)#ip route 217.200.100.0 255.255.255.0 199.150.50.1
Router(config)#ip route 220.200.100.0 255.255.255.0 199.150.50.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#

```

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Figure 9: Static Router Configuration for AB3 Campus

➤ Static Routing Protocol for AB4 Campus Router

```

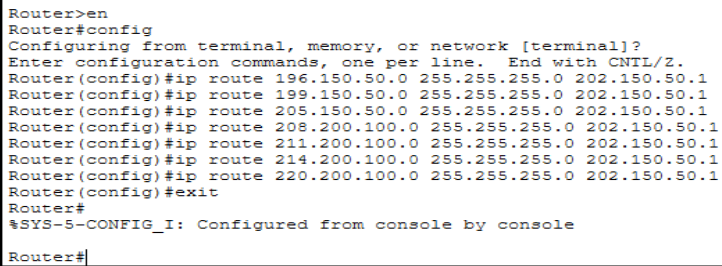
Router>en
Router#config
Router(config)#ip route 196.150.50.0 255.255.255.0 202.150.50.1
Router(config)#ip route 199.150.50.0 255.255.255.0 202.150.50.1

```

```

Router(config)#ip route 205.150.50.0 255.255.255.0 202.150.50.1
Router(config)#ip route 208.200.100.0 255.255.255.0 202.150.50.1
Router(config)#ip route 211.200.100.0 255.255.255.0 202.150.50.1
Router(config)#ip route 214.200.100.0 255.255.255.0 202.150.50.1
Router(config)#ip route 220.200.100.0 255.255.255.0 202.150.50.1
Router(config)#exit

```



```

Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 196.150.50.0 255.255.255.0 202.150.50.1
Router(config)#ip route 199.150.50.0 255.255.255.0 202.150.50.1
Router(config)#ip route 205.150.50.0 255.255.255.0 202.150.50.1
Router(config)#ip route 208.200.100.0 255.255.255.0 202.150.50.1
Router(config)#ip route 211.200.100.0 255.255.255.0 202.150.50.1
Router(config)#ip route 214.200.100.0 255.255.255.0 202.150.50.1
Router(config)#ip route 220.200.100.0 255.255.255.0 202.150.50.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#

```

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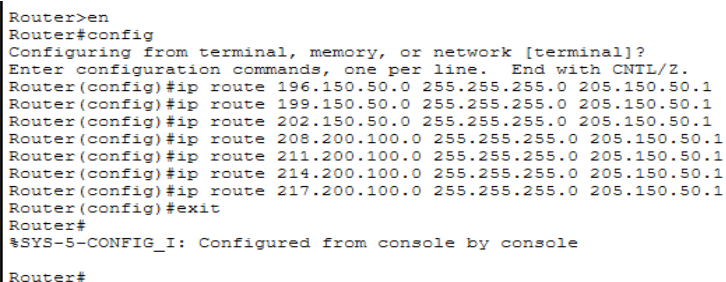
Figure 10: Static Router Configuration for AB4 Campus

➤ Static Routing Protocol for AB5 Campus Router

```

Router>en
Router#config
Router(config)#ip route 196.150.50.0 255.255.255.0 205.150.50.1
Router(config)#ip route 199.150.50.0 255.255.255.0 205.150.50.1
Router(config)#ip route 202.150.50.0 255.255.255.0 205.150.50.1
Router(config)#ip route 208.200.100.0 255.255.255.0 205.150.50.1
Router(config)#ip route 211.200.100.0 255.255.255.0 205.150.50.1
Router(config)#ip route 214.200.100.0 255.255.255.0 205.150.50.1
Router(config)#ip route 217.200.100.0 255.255.255.0 205.150.50.1
Router(config)#exit

```



```

Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 196.150.50.0 255.255.255.0 205.150.50.1
Router(config)#ip route 199.150.50.0 255.255.255.0 205.150.50.1
Router(config)#ip route 202.150.50.0 255.255.255.0 205.150.50.1
Router(config)#ip route 208.200.100.0 255.255.255.0 205.150.50.1
Router(config)#ip route 211.200.100.0 255.255.255.0 205.150.50.1
Router(config)#ip route 214.200.100.0 255.255.255.0 205.150.50.1
Router(config)#ip route 217.200.100.0 255.255.255.0 205.150.50.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#

```

Copy

Paste

Figure 11: Static Router Configuration for AB5 Campus

2.5 DHCP Server Configuration

Each campus contains a dedicated DHCP server connected to the campus switch. The DHCP server is responsible for automatically assigning IP addresses to hosts within its respective campus LAN.

Each DHCP server was configured with the following parameters:

- Network address and subnet mask
- Default gateway (router LAN interface)
- DNS server IP address (Campus AB1 DNS Server)

All campus PCs were configured to obtain IP addresses automatically using DHCP. This distributed DHCP design reduces manual configuration, improves reliability, and ensures efficient IP address management within each campus.

➤ Campus AB1 DHCP

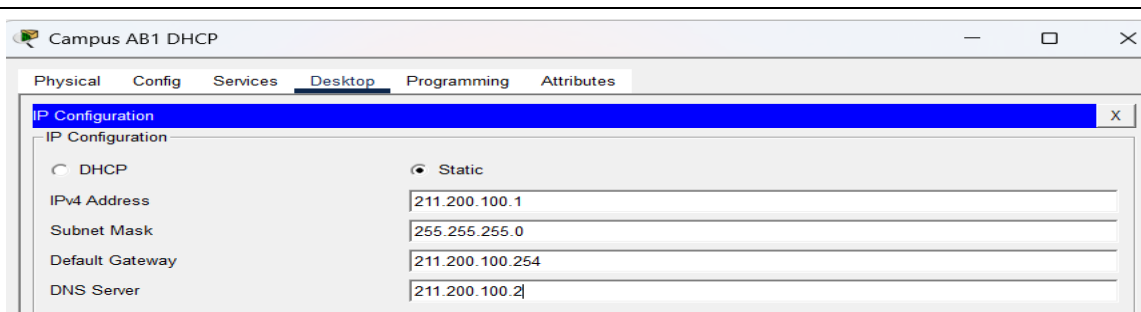


Figure 12: DHCP Configuration for Campus AB1

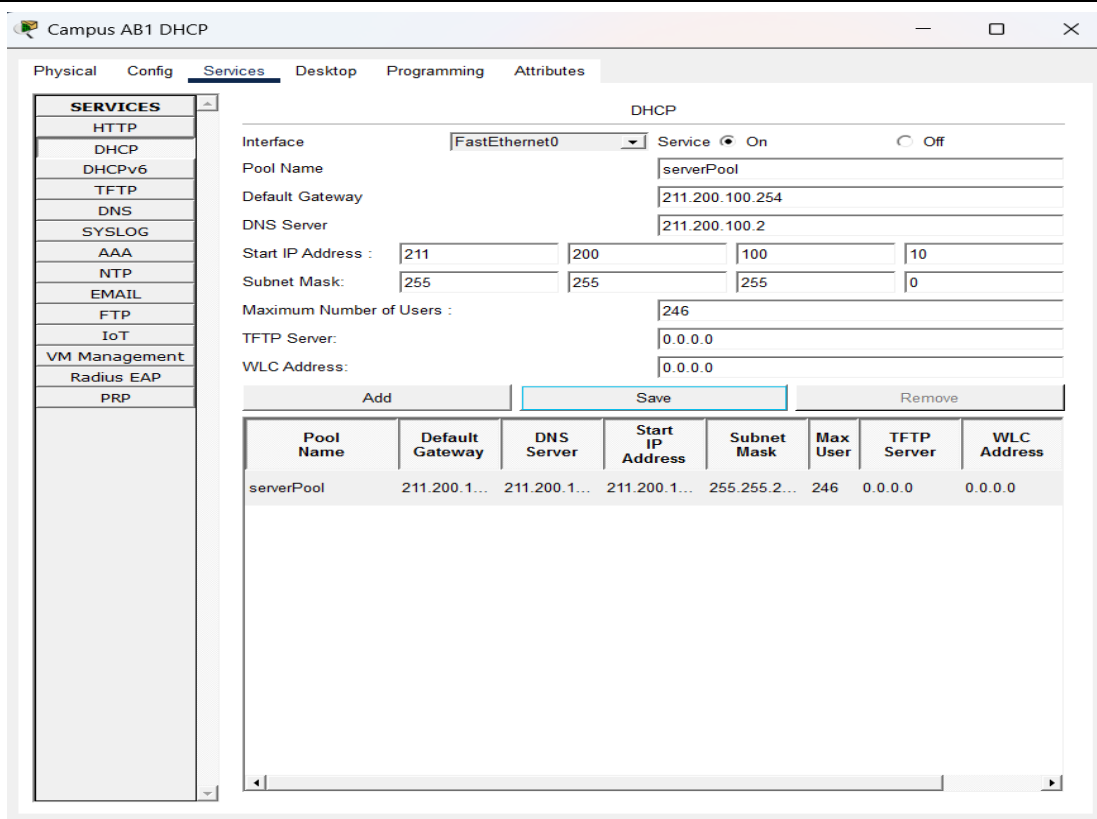


Figure 13: DHCP Service Running for AB1

➤ Automatic IP Assignment to PCs in AB1

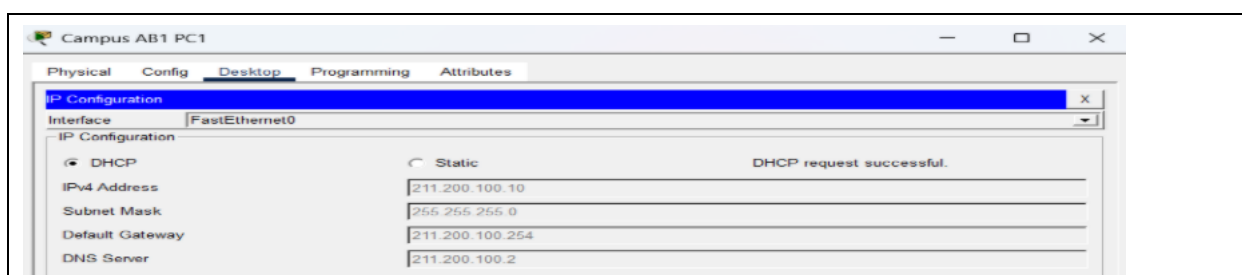


Figure 14.1: Automatic DHCP IP Assignment to PC1 in AB1

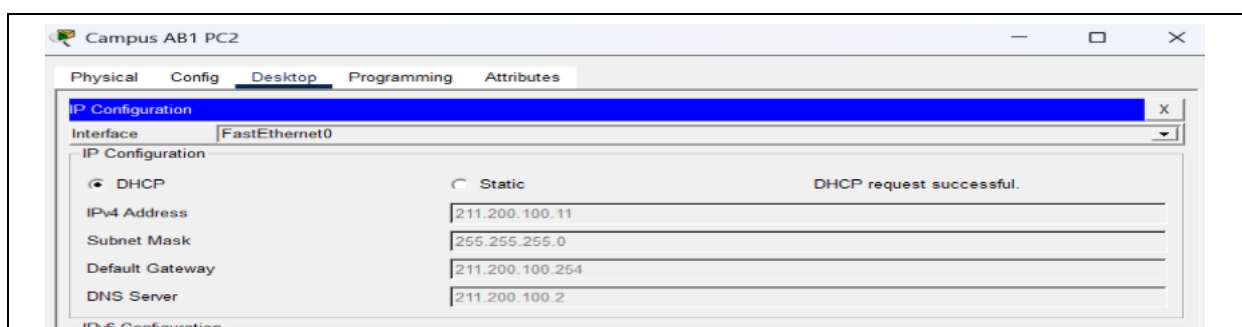


Figure 14.2: Automatic DHCP IP Assignment to PC2 in AB1

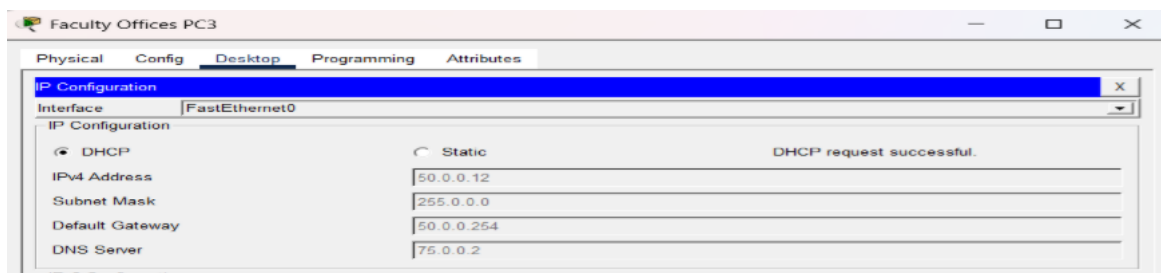


Figure 14.3: Automatic DHCP IP Assignment to PC3 in AB1

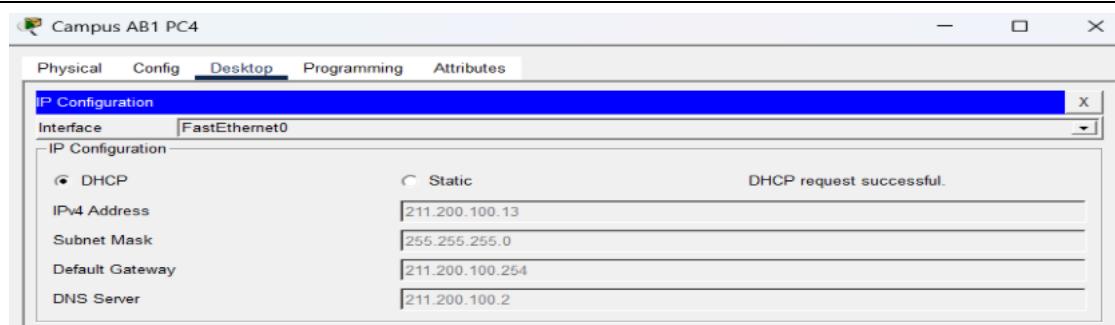


Figure 14.4: Automatic DHCP IP Assignment to PC4 in AB1

➤ Campus AB2 DHCP

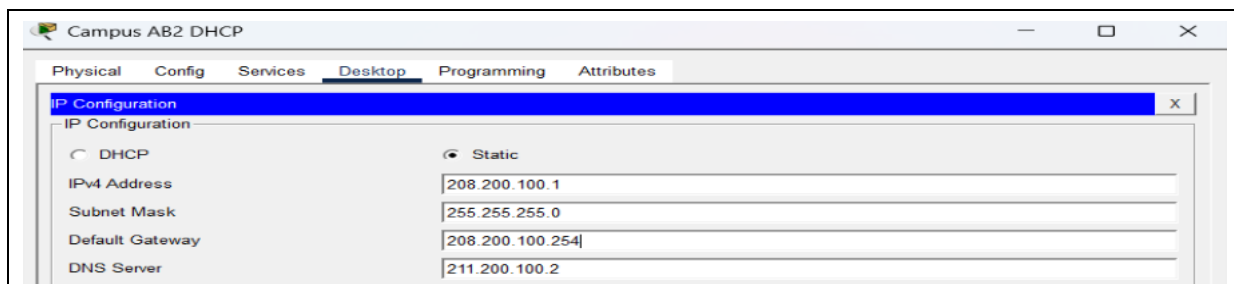


Figure 15: DHCP Configuration for Campus AB2

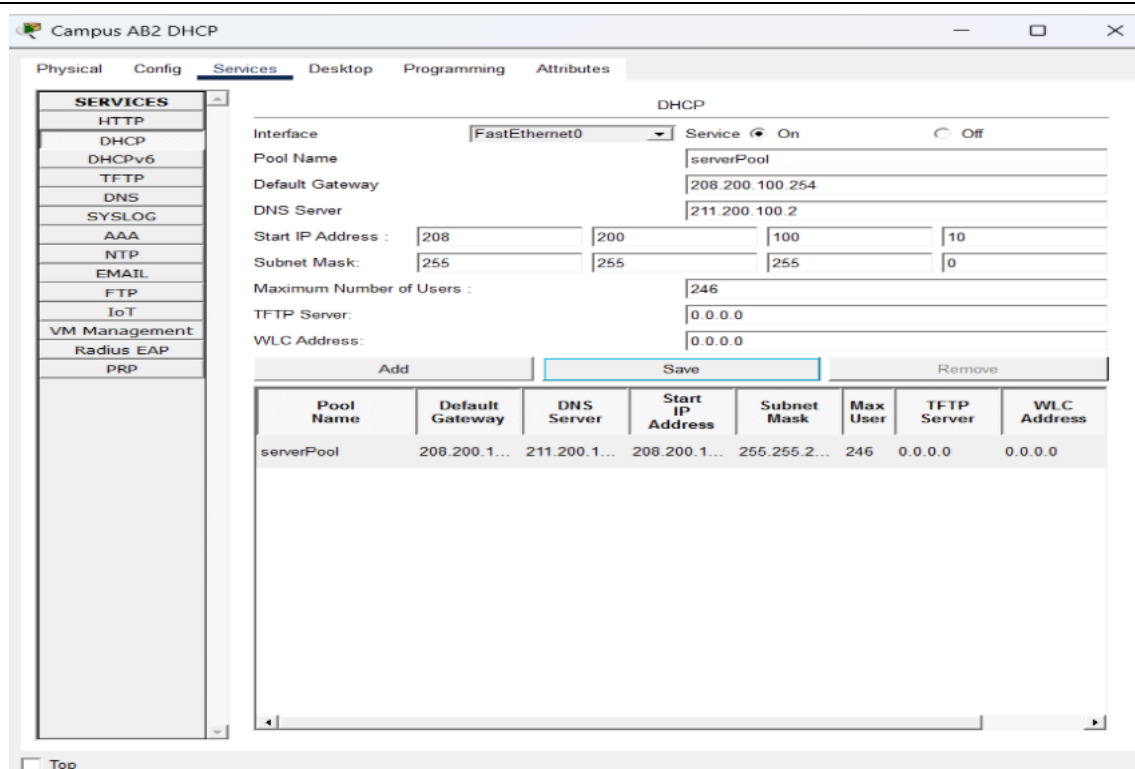


Figure 16: DHCP Service Running for AB2

➤ Automatic IP Assignment to PCs in AB2

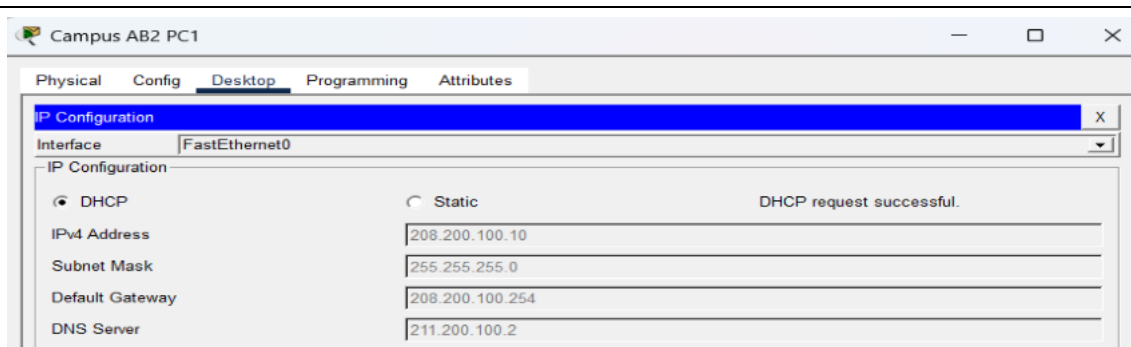
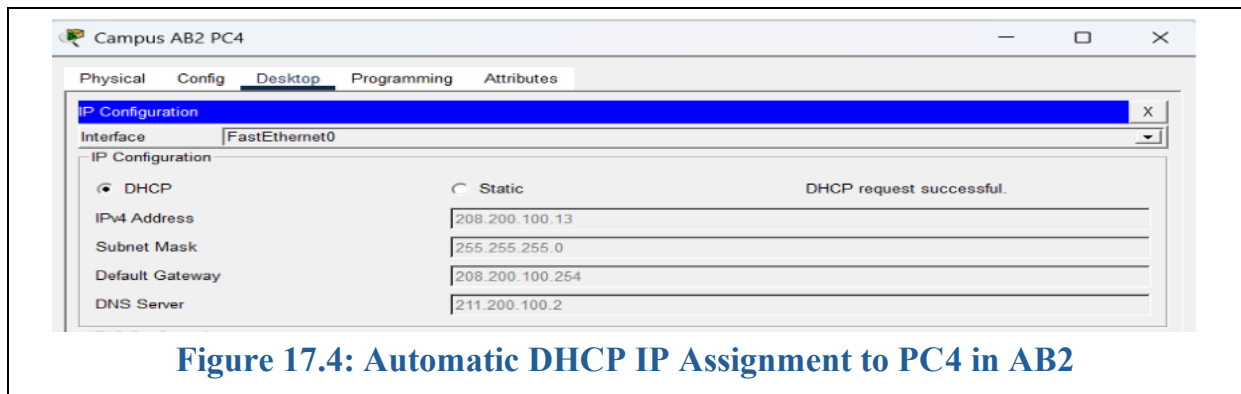
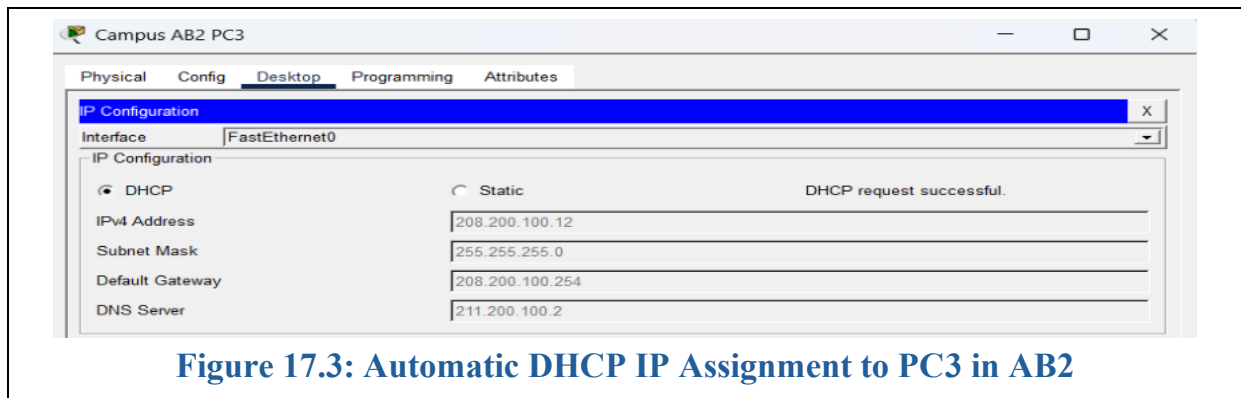
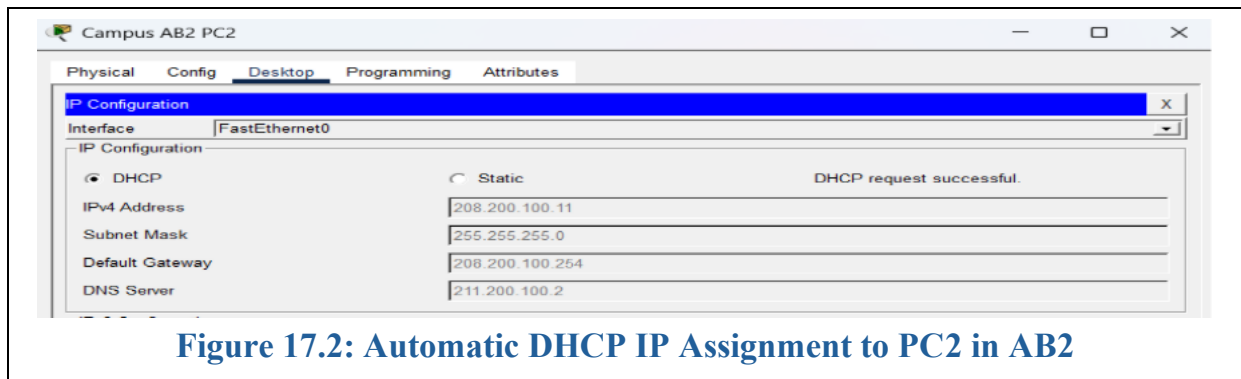


Figure 17.1: Automatic DHCP IP Assignment to PC1 in AB2



➤ Campus AB3 DHCP

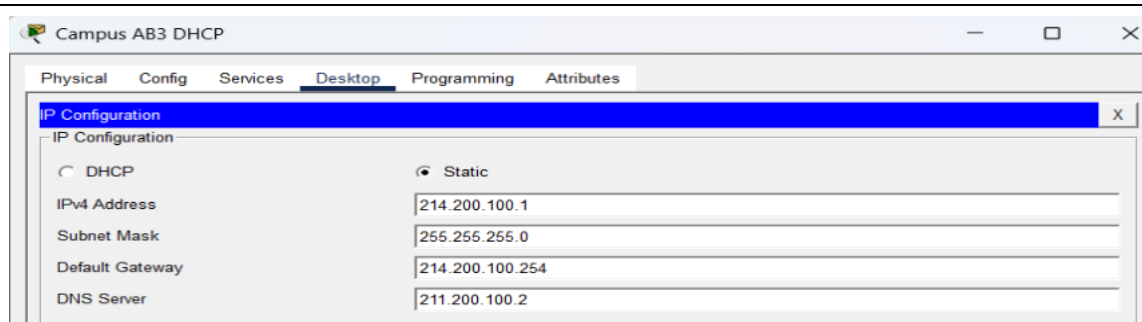


Figure 18: DHCP Configuration for AB 3

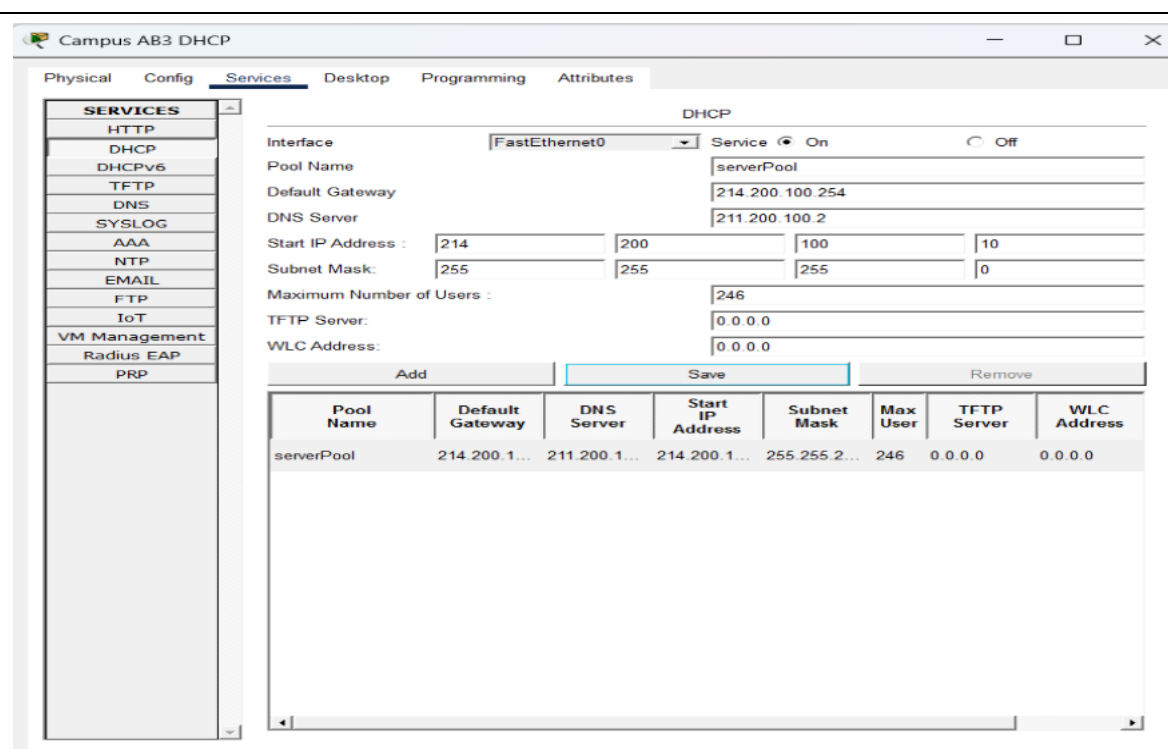


Figure 19: DHCP Service Running for AB3

➤ Automatic IP Assignment to PCs in AB3

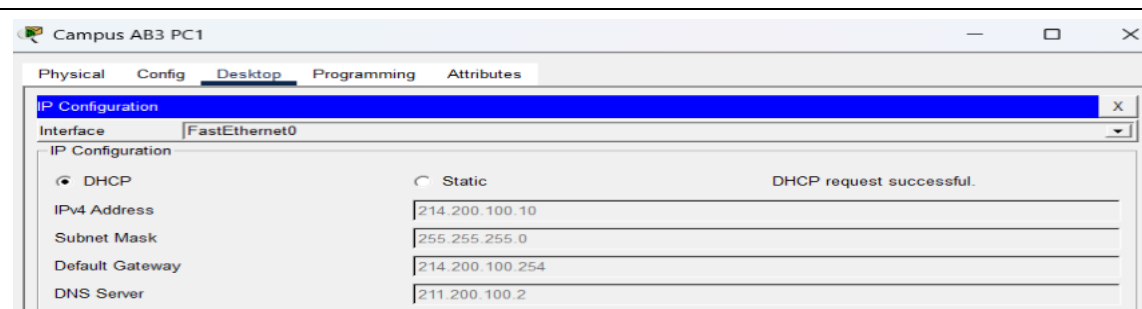


Figure 20.1: Automatic DHCP IP Assignment to PC1 in AB3

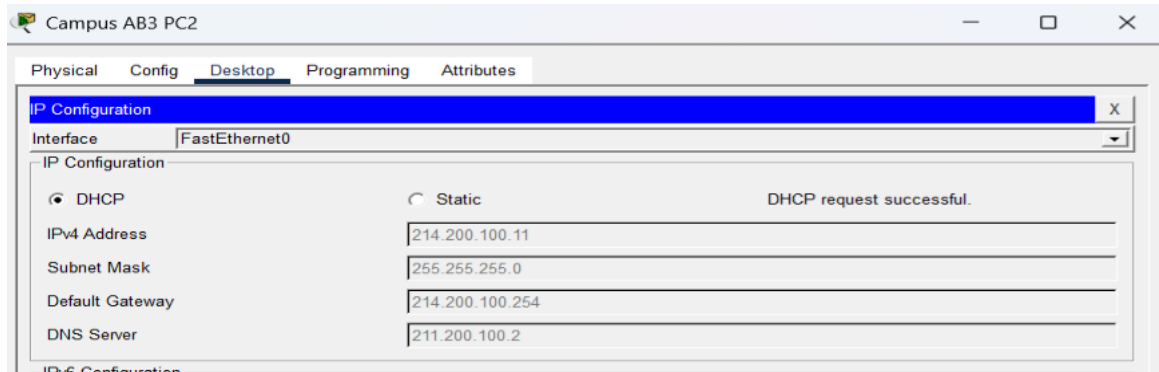


Figure 20.2: Automatic DHCP IP Assignment to PC2 in AB3

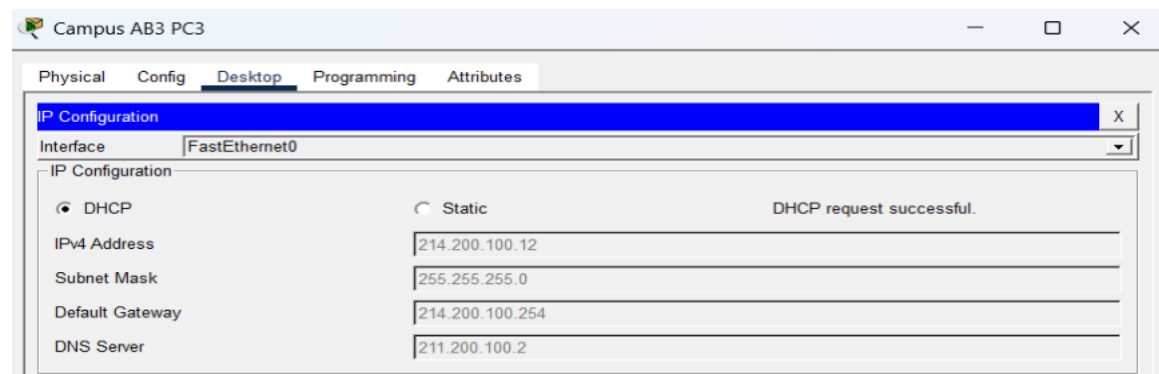


Figure 20.3: Automatic DHCP IP Assignment to PC3 in AB3

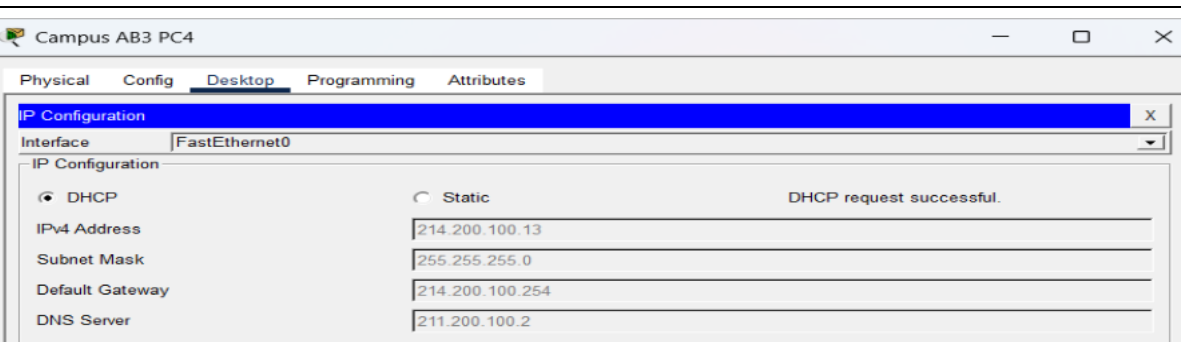
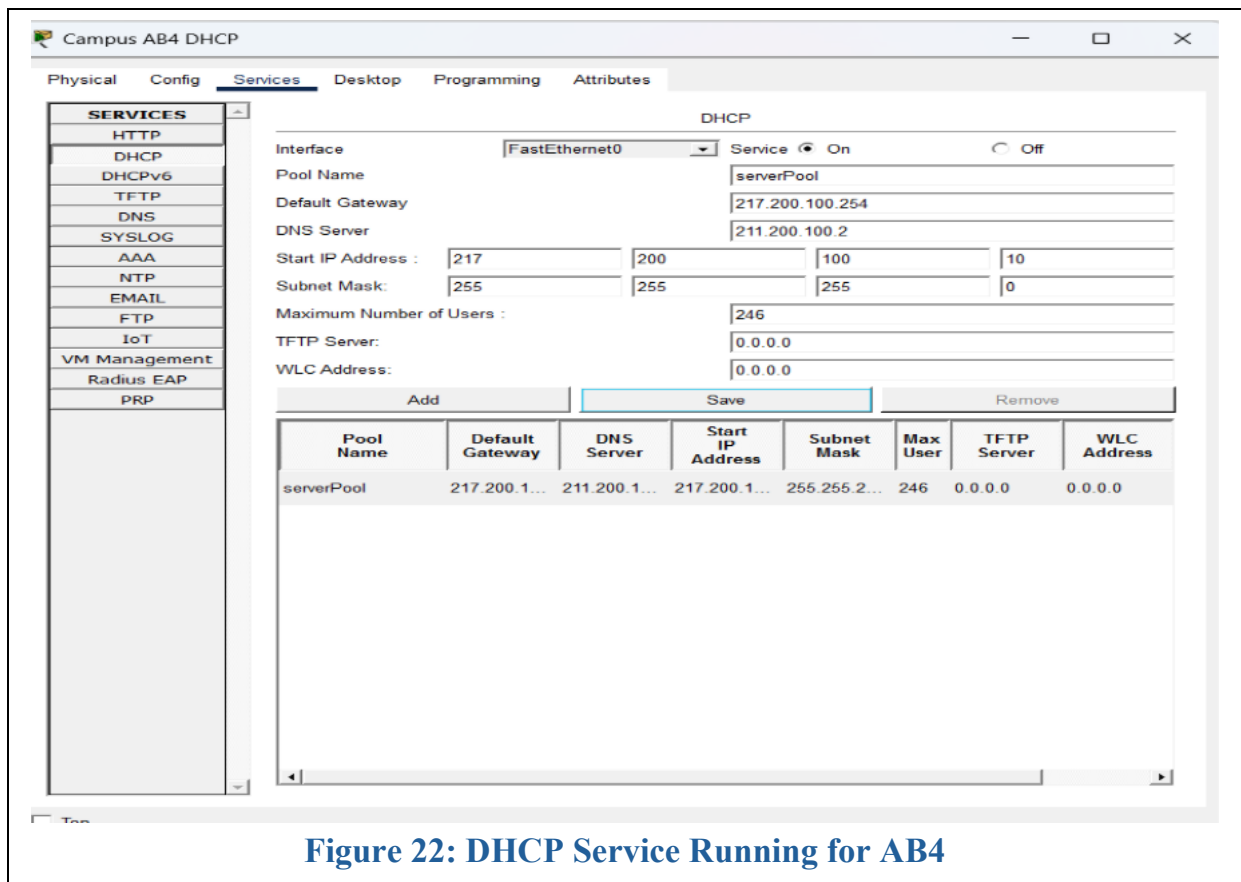
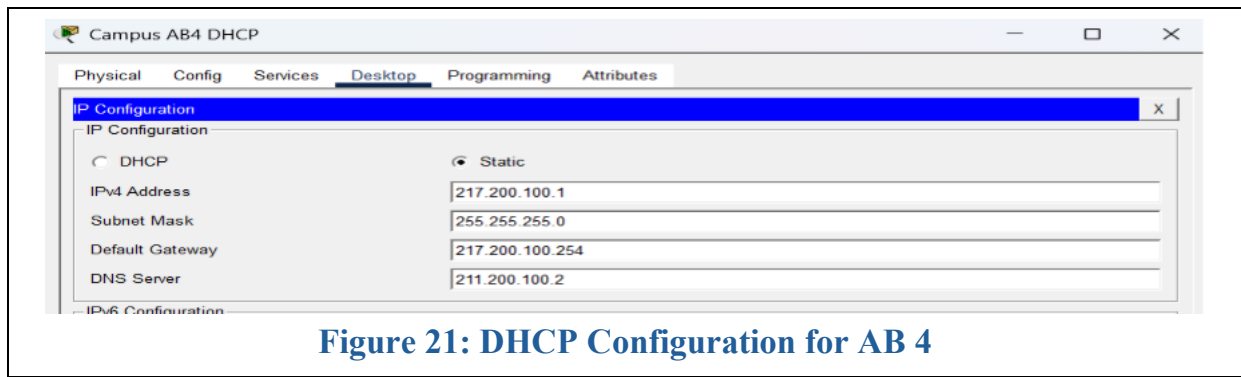
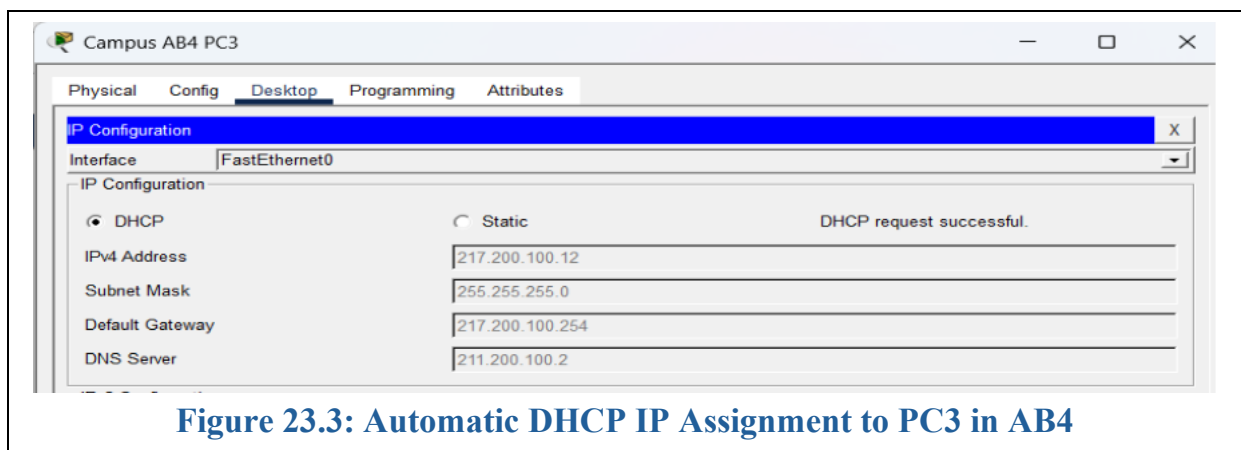
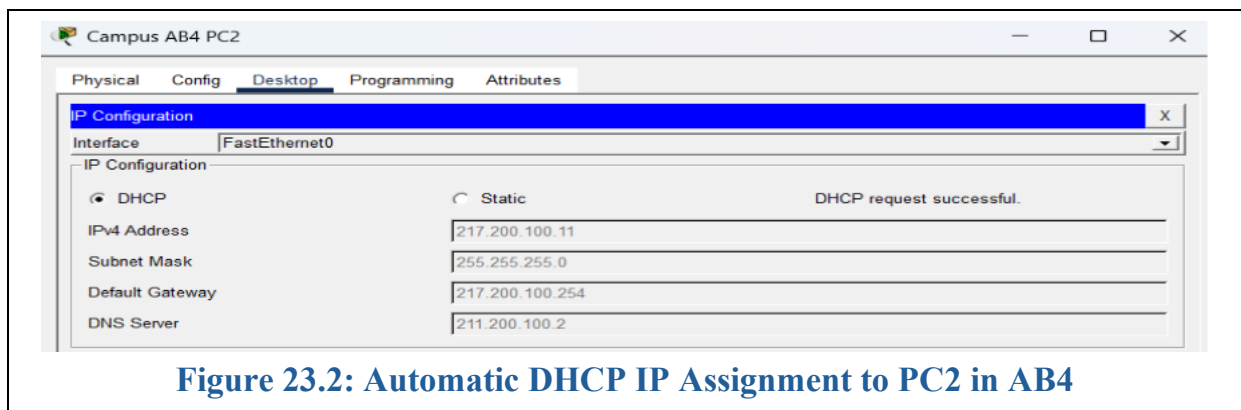
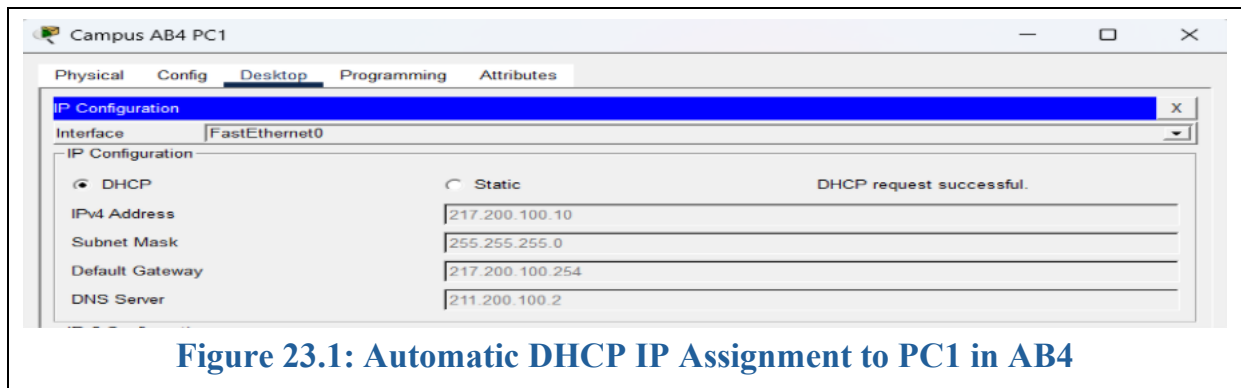


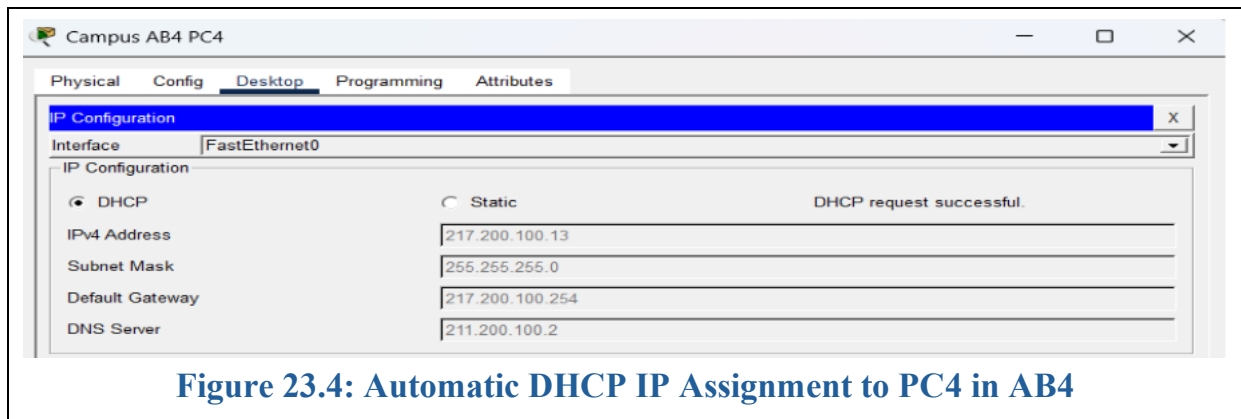
Figure 20.4: Automatic DHCP IP Assignment to PC4 in AB3

➤ Campus AB4 DHCP

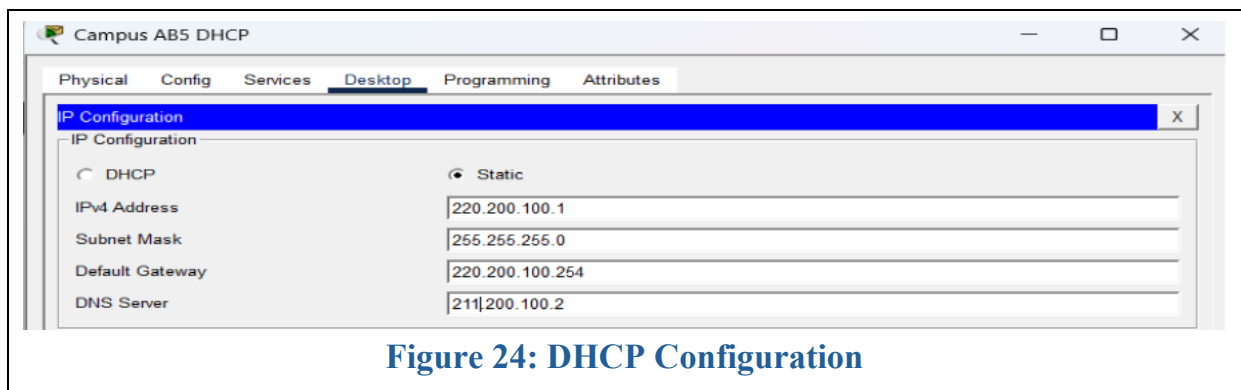


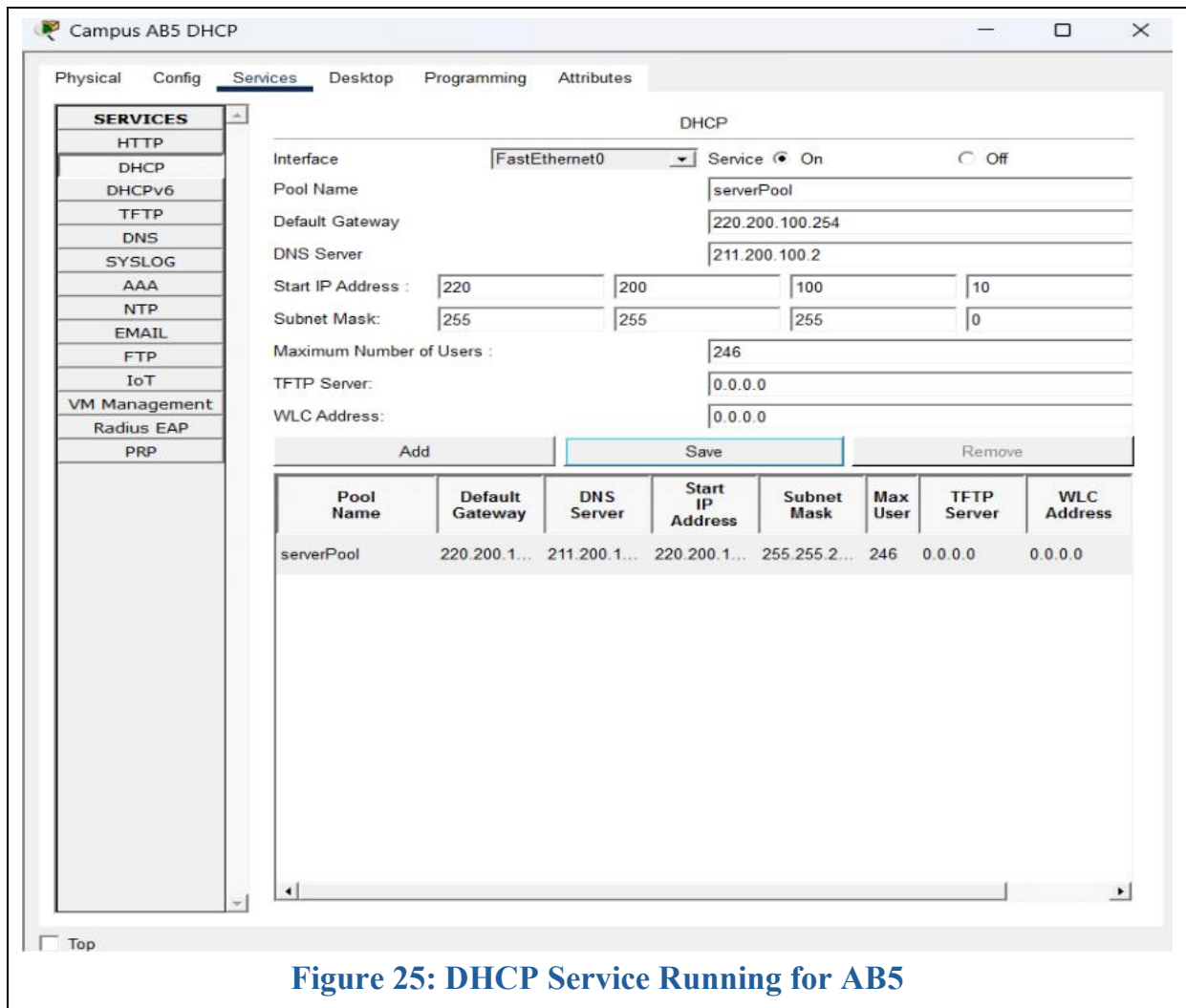
➤ Automatic IP Assignment to PCs in AB4



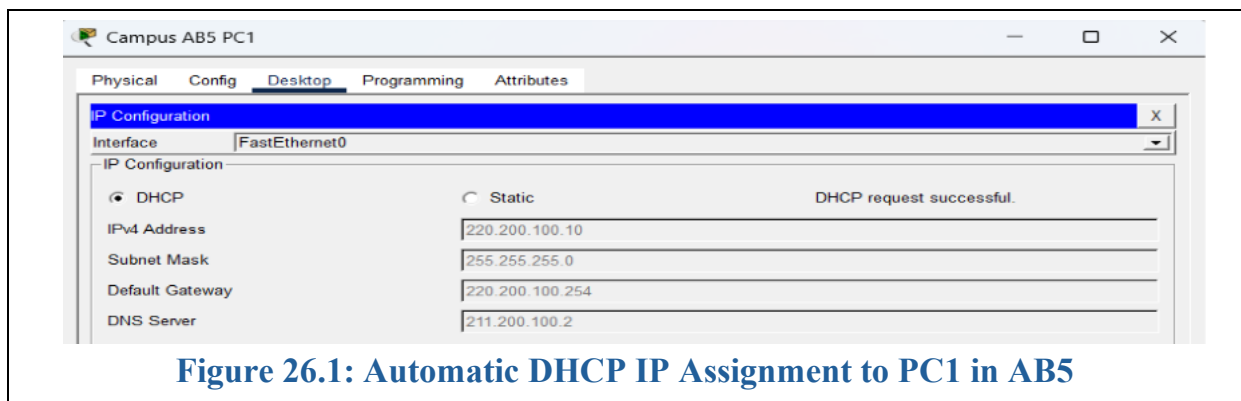


➤ Campus AB5 DHCP





➤ Automatic IP Assignment to PCs in AB5



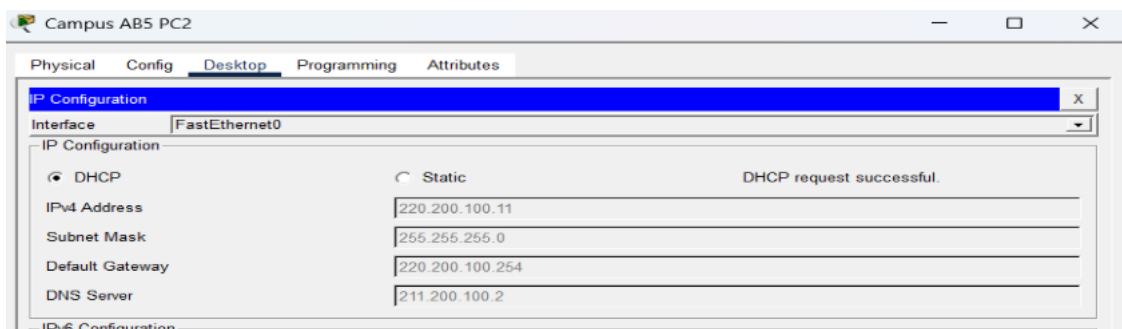


Figure 26.2: Automatic DHCP IP Assignment to PC2 in AB5

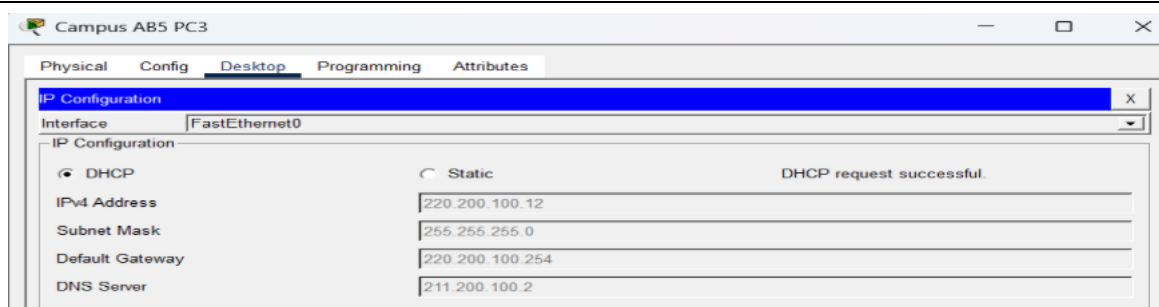


Figure 26.3: Automatic DHCP IP Assignment to PC3 in AB5

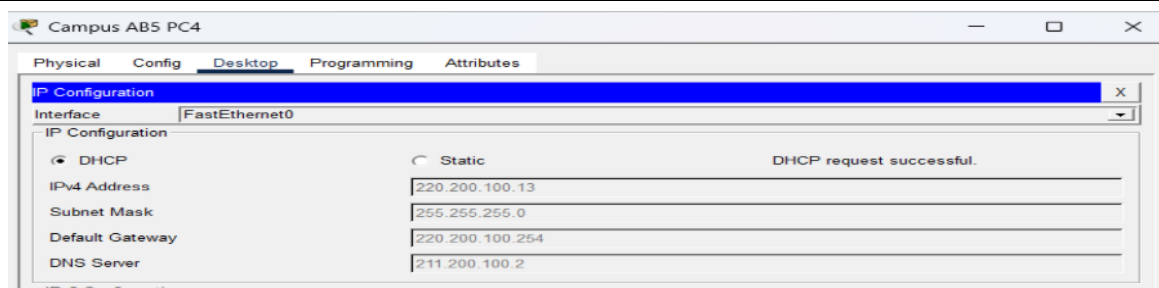


Figure 26.4: Automatic DHCP IP Assignment to PC4 in AB5

2.6 DNS Server Configuration

After completing the static routing configuration and setting up the DHCP server, a centralized DNS server was configured in Campus AB1. This DNS server is responsible for resolving domain names to IP addresses within the campus network and has the IP 211.200.100.2.

The domain <http://www.university.edu.bd> was created and mapped to the IP address of the web server. Successful DNS resolution allows users to access network services using domain names instead of IP addresses.

The screenshot shows the 'Campus AB1 DNS' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing 'Static' IP settings. The 'IPv6 Configuration' section is also visible, showing 'Static' settings. The '802.1X' section is at the bottom with 'Use 802.1X Security' unchecked.

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	211.200.100.2
Subnet Mask	255.255.255.0
Default Gateway	211.200.100.254
DNS Server	211.200.100.2

IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::201:64FF:FE86:19A8
Default Gateway	
DNS Server	

802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

Figure 27: DNS Server Configuration in AB1

The screenshot shows the 'Campus AB1 DNS' configuration window with the 'Services' tab selected. The 'DNS' service is listed in the 'SERVICES' sidebar and is currently 'On'. The 'Resource Records' section shows a single record for 'www.university.edu.bd' with type 'A Record' and address '211.200.100.3'.

DNS			
DNS Service		☒ On	☐ Off
Resource Records			
Name	www.university.edu.bd		Type A Record
Address	211.200.100.3		
Add		Save	Remove
No.	Name	Type	Detail
0	www.university.edu.bd	A Record	211.200.100.3

DNS Cache			

Figure 28: DHCP Service Running for AB1

2.7 Web Server Configuration

Following the DNS configuration, a web server was set up in Campus AB1 to host a webpage accessible across the entire campus network. The domain <http://www.university.edu.bd> was mapped to the university's web server (211.200.100.3). The webpage displays my name, student ID, and a small amount of personal information, as required by the project instructions.

The webpage was successfully accessed from multiple facilities using a standard web browser, confirming proper HTTP service operation and end to end connectivity. This setup allows users across all campuses to access the university website using the domain name instead of the IP address, providing personalized access.

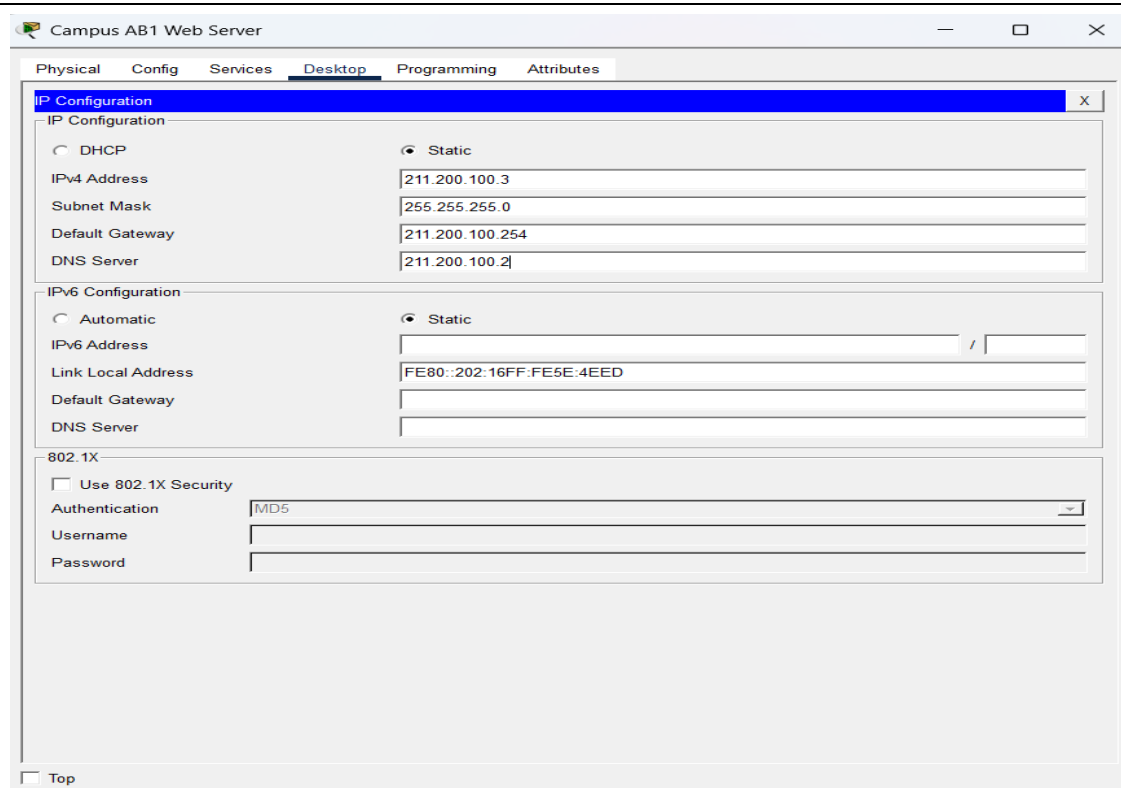


Figure 29: Web Server Configuration in AB1

3. Experimental Results

3.1 Ping Test Results

Ping commands were executed between hosts of different campuses (AB1, AB2, AB3, and AB4). Successful ICMP replies confirmed:

- Correct IP addressing
- Proper static routing
- End to end connectivity

➤ Ping Command

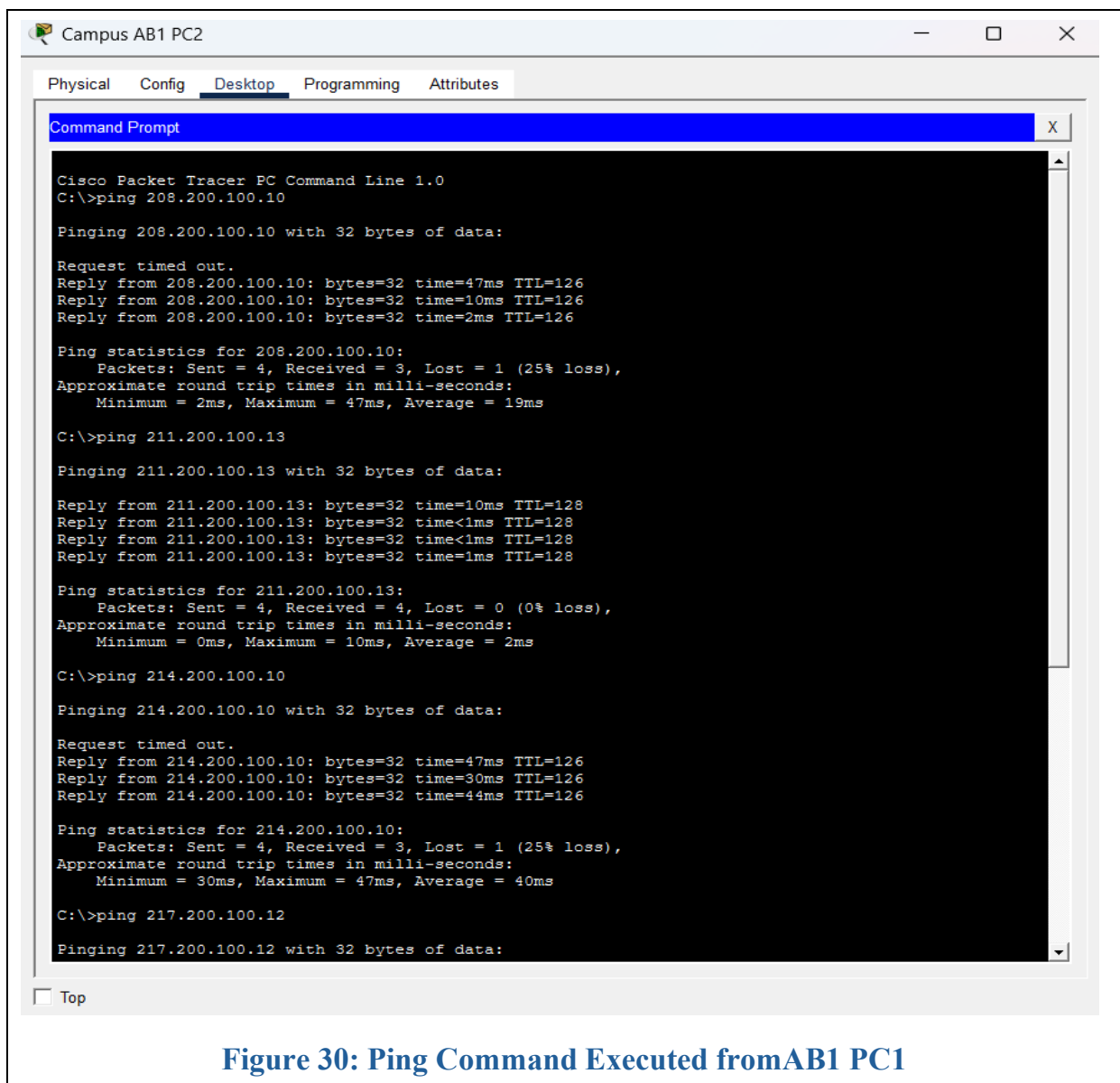


Figure 30: Ping Command Executed from AB1 PC1

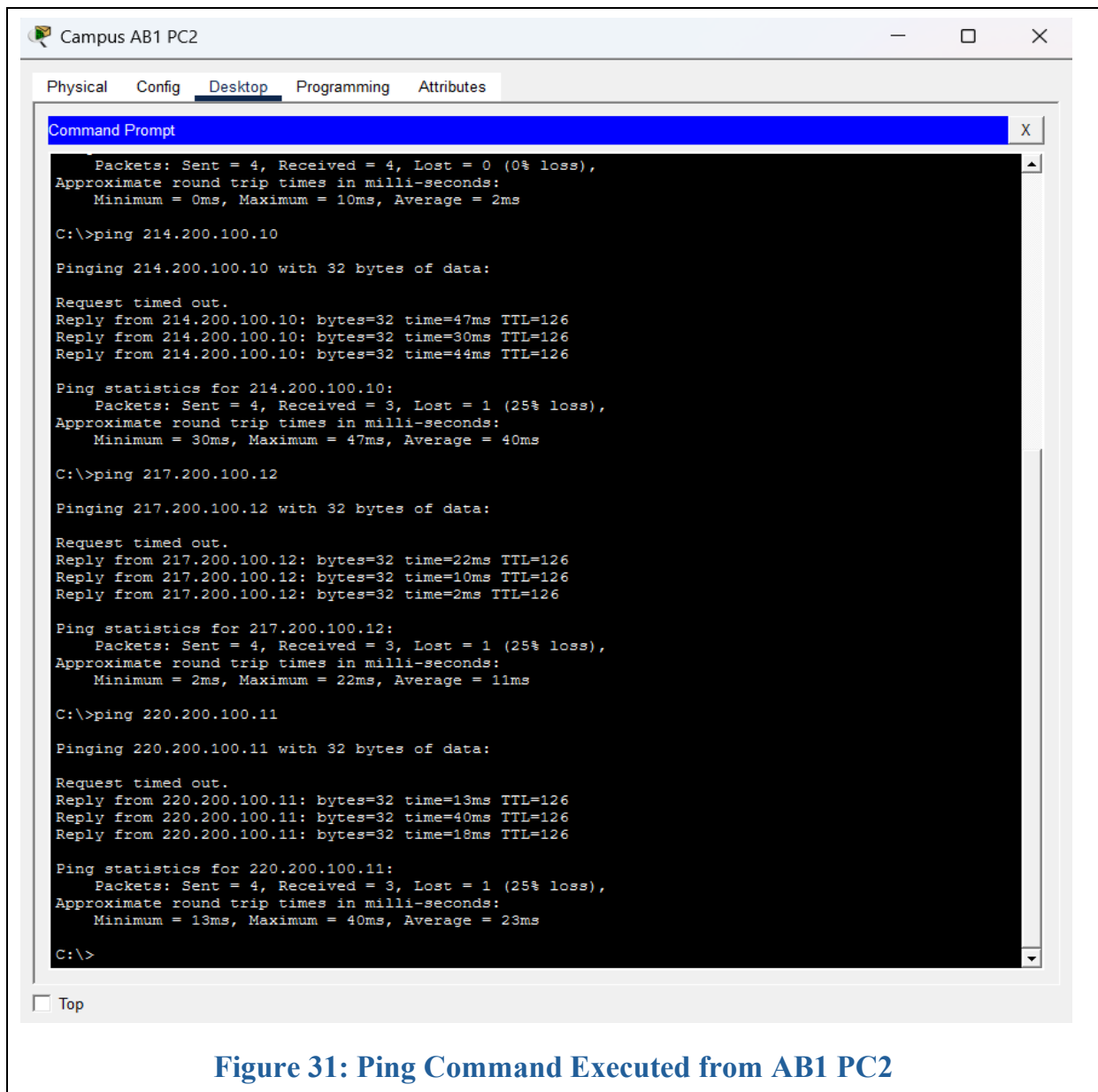
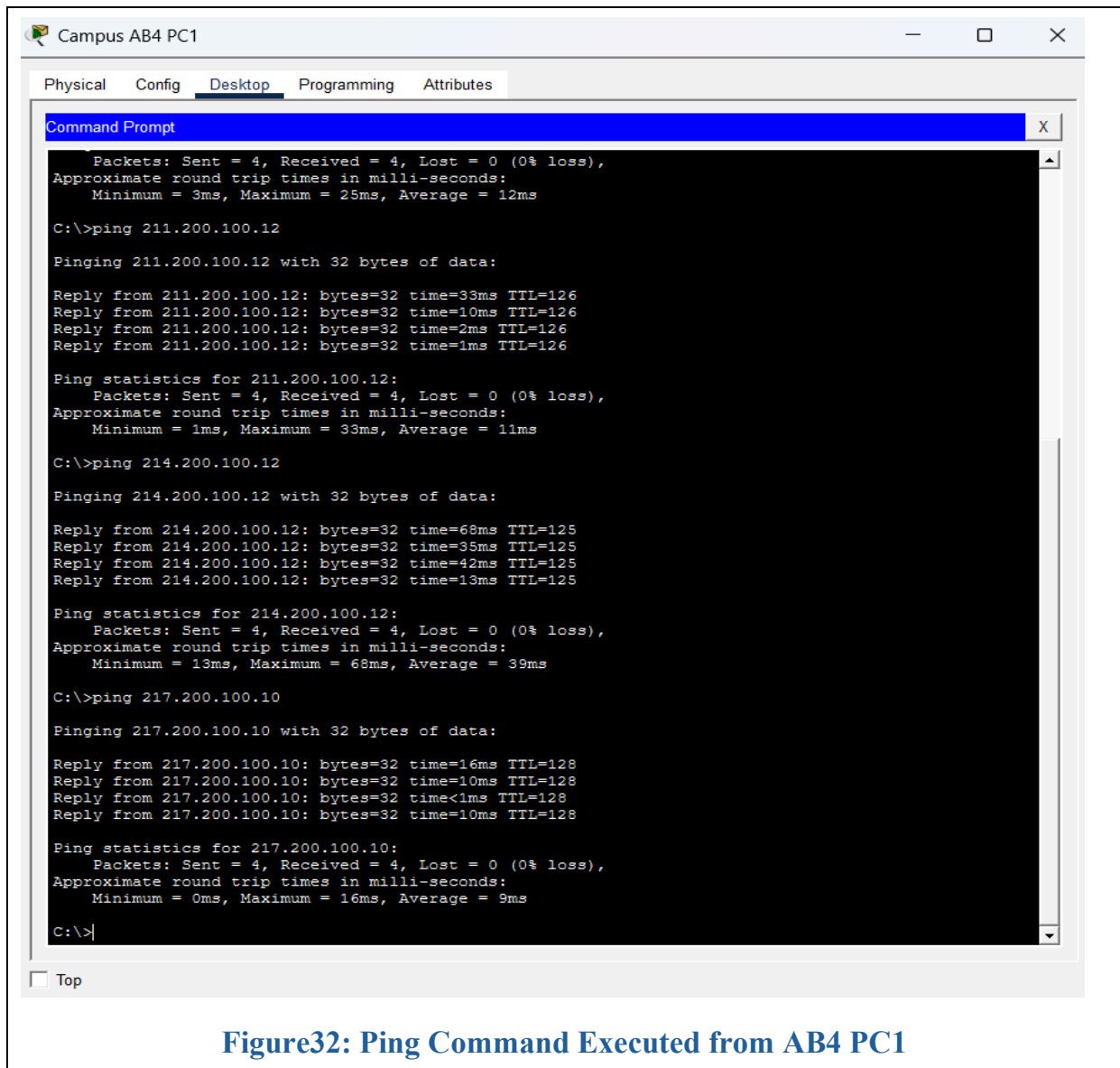
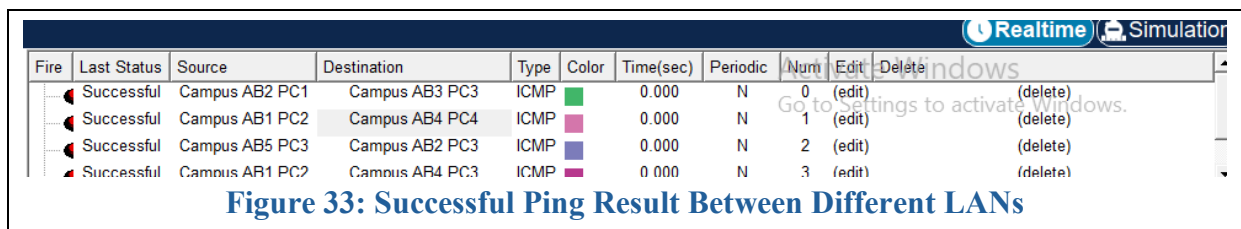


Figure 31: Ping Command Executed from AB1 PC2

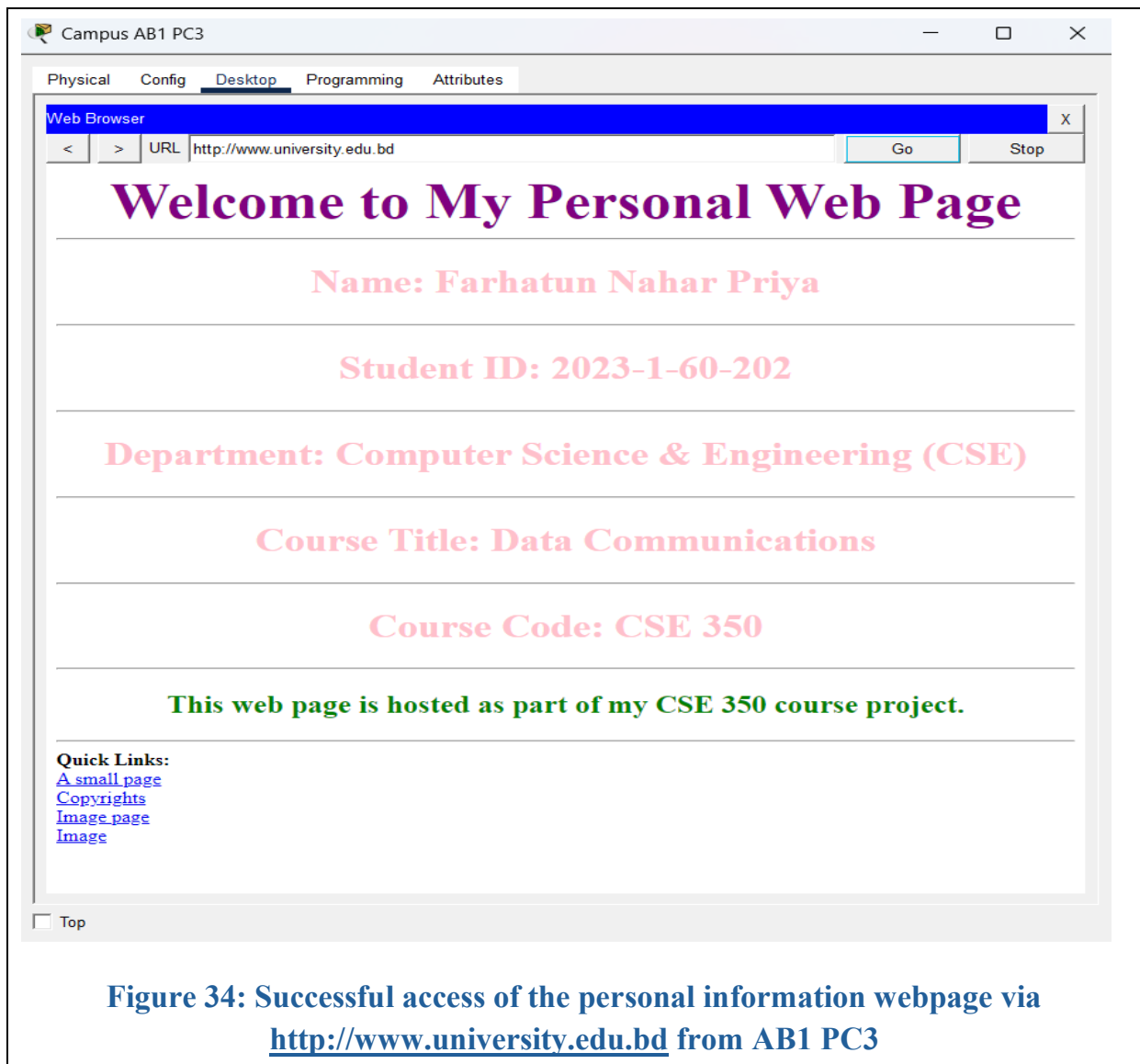


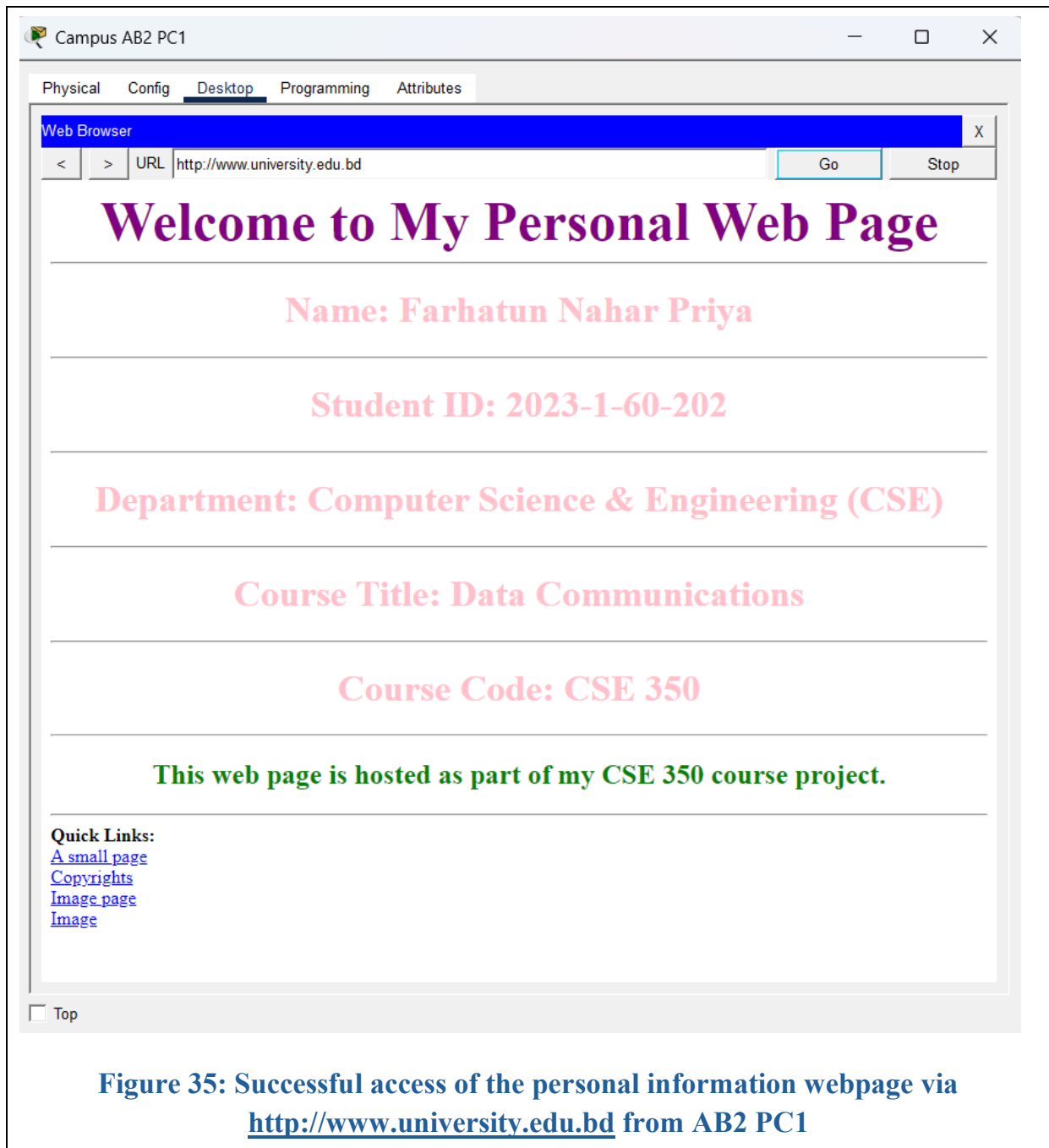
➤ Ping Test Result



3.2 Web Access Verification

The domain <http://www.university.edu.bd> was accessed from multiple PCs located in different Campuses. The webpage loaded successfully in all cases, verifying proper DNS resolution and web server functionality across the campus network.











4. Conclusion

This project successfully implements a university network across five campuses: AB1, AB2, AB3, AB4 and AB5, using Class C IP addressing and static routing. Each campus operates as an independent LAN while maintaining full connectivity with the central network. Distributed DHCP servers provide automated IP allocation for each campus, and centralized DNS and web services ensure seamless access to online resources.

The network design is efficient, reliable and easy to manage, fulfilling all specified project requirements. Furthermore, the architecture is scalable, making it suitable for both academic and administrative purposes. Overall, the project demonstrates a practical application of networking principles in designing a realistic and functional university network infrastructure.